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Enhancing stock market price prediction through FinGPT-driven sentiment analysis, technical indicators, historical price data, and attention-based hybrid deep learning models

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Accurate stock price forecasting is increasingly dependent on models that can integrate heterogeneous market signals beyond traditional quantitative indicators. Classical sentiment analysis tools and lexicon-based methods often fail to capture the contextual and domain-specific nuances of financial language, limiting their predictive contribution. To address these limitations, this study introduces a multimodal forecasting framework that leverages FinGPT, an advanced financial large language model specifically trained for market-related text understanding. FinGPT is employed to extract high-resolution sentiment polarity scores from Twitter data, replacing conventional sentiment models and providing a more robust representation of investor psychology. These sentiment features are fused with historical stock prices and technical indicators to form a comprehensive input space. The integrated dataset is used to evaluate four deep learning architectures: CNN, LSTM, Attention-based LSTM (ALSTM), and a newly proposed Hybrid ALSTM-CNN model with an attention mechanism. Experiments performed on eight major NASDAQ and NYSE stocks demonstrate that incorporating FinGPT-derived sentiment significantly enhances forecasting accuracy across all models. The Hybrid model consistently achieves the best performance, effectively capturing both long-term dependencies and short-term market dynamics. Results confirm that FinGPT-driven sentiment modeling provides substantial predictive value compared with traditional sentiment approaches, enabling more realistic, behavior-aware stock price forecasting. For instance, in the case of AMD stock, it recorded the lowest MSE (46.06) and the highest R^2 (0.624), highlighting its superior ability to capture nonlinear dependencies and market dynamics. Overall, the proposed sentiment-enhanced hybrid framework provides a more robust and realistic procedure for stock price prediction by integrating quantitative and qualitative market signals.

Keywords Stock price prediction, Sentiment analysis, FinGPT, Deep learning, Technical analysis, Multimodal data fusion

Abbreviations

DL	Deep Learning
CNN	Convolutional Neural Networks
MAE	Mean Absolute Error
MSE	Mean Squared Error
RNN	Recurrent Neural Networks
LSTM	Long Short-Term Memory
SVR	Support Vector Regression
NLP	Natural Language Processing

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DNN	Deep Neural Networks
Bi-LSTM	Bidirectional LSTM
RBF	Radial Basis Function
RF	Random Forests
ANN	Artificial Neural Networks
MV	Mean-Variance
ML	Machine Learning
DTR	Decision Tree Regression
TEMA	Triple Exponential Moving Average
VMD	Variational Mode Decomposition
TLSTM	Transductive Long Short-Term Memory
DJIA	Dow Jones Industrial Average
ARIMA	Autoregressive Integrated Moving Average
WMA	Weighted Moving Average
LASSO	Least Absolute Shrinkage and Selection Operator
TFJ-DRL	Time-driven Feature-aware Jointly Deep Reinforcement Learning
SA	Sentiment Analysis
AMD	Advanced Micro Devices, Inc
AMZN	Amazon.com, Inc
AAPL	Apple Inc
GOOG	Alphabet Inc
ReLU	Rectified Linear Unit
SVM	Support Vector Machine
AI	Artificial Intelligence
MFI	Money Flow Index
RS	Relative Strength
RSI	Relative Strength Index
MA	Moving Average
MACD	Moving Average Convergence Divergence
EMA	Exponential Moving Average
GRU	Gated Recurrent Unit
RMSE	Root Mean Squared Error
MAPE	Mean Absolute Percentage Error
GAP	Global Average Pooling
LR	Logistic Regression
SMA	Simple Moving Average
KNN	K-Nearest Neighbors
TI	Technical Indicator
DRL	Deep Reinforcement Learning
PPO	Proximal Policy Optimization
LLM	Large Language Model
DPS	Daily Polarity Score
VADER	Valence Aware Dictionary and sEntiment Reasoner
XGBoost	Extreme Gradient Boosting
FinGPT	Finance Generative Pre-trained Transformer
FinBERT	Finance Bidirectional Encoder Representations from Transformers
XAI	Explainable Artificial Intelligence
MSFT	Microsoft Corporation
PG	Procter & Gamble Co
TSLA	Tesla, Inc
TSM	Taiwan Semiconductor Manufacturing Company Limited

Precise stock price forecasting serves as a vital tool for investors, enabling them to make well-informed decisions, recognize profitable opportunities, and mitigate financial risks. Anticipating market trends and price fluctuations allows investors to design effective trading strategies and enhance portfolio performance amid dynamic and volatile financial conditions. AI has profoundly transformed the landscape of time series anticipation, notably in financial domains such as stock market prediction. Traditional econometric models including ARIMA are often limited by their linear assumptions and inability to adapt to the nonlinear, chaotic, and highly volatile nature of financial data¹. In contrast, AI-based models such as ANNs, DL architectures, and metaheuristic-optimized systems are capable of learning complex nonlinear relationships and temporal dependencies directly from data without requiring strong prior assumptions about distributional properties. Stock price prediction has remained a central challenge in computational finance because of the inherently complex and nonlinear behavior of financial markets. Models relying solely on historical price information often fail to account for behavioral and informational factors that drive market dynamics.

Researchers have demonstrated that the sentiment polarity derived from tweets or financial news exhibits a measurable correlation with short-term market fluctuations, indicating that investors' collective emotions often precede price trends. In recent years SA has developed as a critical complementary approach leveraging textual data such as financial news and social media posts to capture public mood and investor perception, which significantly affect stock price movements^{2,3}. In the past, SA in financial studies primarily relied on lexicon-based

approaches and traditional LSTM-based models to interpret textual data. However, with recent advancements in DNN and artificial intelligence, modern research increasingly employs transformer-based architectures such as BERT, FinBERT, and FinGPT, which offer superior contextual understanding and domain-specific sentiment interpretation in financial text analysis. A FinBERT-LSTM framework was developed to integrate financial news sentiment into a DL model, demonstrating that textual sentiment extracted from market-, industry-, and stock-related news can enhance forecasting accuracy beyond models relying solely on price data^{4,5}. Likewise, it has been established that aggregated Twitter sentiment exhibits a significant correlation with the DJIA, indicating that social media information can serve as a valuable proxy for investor psychology^{6,7}.

Building upon these insights, a growing body of research emphasizes the integration of multi-source or multimodal data to improve the robustness and interpretability of forecasting models. Rather than relying on a single information stream, contemporary approaches combine historical price data, technical indicators, and textual sentiment into unified predictive systems. The multimodal design reflects the complex interplay between market fundamentals, technical patterns, and behavioral sentiment cues⁸. Studies highlight that the fusion of structured numerical data with unstructured textual content leads to richer feature representations and improved learning performance⁹. A hybrid RNN architecture combining GRU, LSTM, and Bi-LSTM layers is introduced to improve stock market prediction through the integration of numerical price data and sentiment extracted from financial news. It is shown that the incorporation of VADER-based sentiment scores significantly enhances forecasting accuracy, leading to lower MAE and RMSE compared with standalone LSTM, GRU, or traditional RNN models¹⁰. Studies underline the importance of multi-source learning, where the combination of technical and textual signals enables models to better reflect real-world market behavior.

However, the persistent uncertainty surrounding the predictability of financial markets, together with an overreliance on historical observations and the application of parameters estimated from past data to real-world decision-making, continues to pose significant challenges for investors^{11,12}. Significant advancements have been achieved in ML and DL frameworks for predicting financial outcomes. Early studies relied on statistical and shallow learning techniques such as SVM, RF, and regression models¹³. However, the nonlinearity and volatility of stock price dynamics necessitated the development of DL architectures capable of learning hierarchical temporal patterns. RNNs, particularly LSTM and GRU variants, have proven effective for modeling sequential dependencies in time-series data. The integration of convolutional layers with LSTM networks has further enhanced model performance by enabling simultaneous learning of temporal and spatial patterns in multivariate financial inputs. Hybrid models such as CNN-LSTM, ALSTM, and HRNN have been found to outperform traditional models by detecting both short-term fluctuations and long-term dependencies in stock prices¹⁴. Moreover, the combination of DL with sentiment-aware modules such as FinBERT-LSTM and hybrid RNNs has yielded superior forecasting accuracy by embedding semantic understanding of financial narratives within temporal models^{4,15,16}.

In light of this evolving literature, the present study contributes to the field by developing an integrated DL framework that leverages three distinct data modalities: historical prices, technical indicators, and textual sentiment scores to forecast stock prices. The proposed approach aims to demonstrate that the inclusion of sentiment polarity, computed from FinGPT-based analysis of tweets, enhances predictive precision and stability across multiple DL architectures, particularly in the hybrid ALSTM-CNN model. Through this integration of quantitative and qualitative information, the research seeks to provide empirical evidence supporting the hypothesis that models incorporating both market behavior and investor sentiment outperform those relying solely on numerical data.

The remainder of this paper is structured in the following manner: section “[Literature review](#)” provides a comprehensive overview of the related literature, highlighting key findings, limitations, and research gaps in stock price forecasting and SA. Section “[Case study and data collection](#)” offers an in-depth explanation of the case study and the datasets utilized, including the datasets used, data preprocessing steps, and feature extraction procedures. Section “[Methodology](#)” outlines the proposed methodology, elaborating on the structure of the predictive models and the integration of sentiment data into the forecasting framework. Section “[Experiment and results](#)” presents the experimental design and outcomes, comparing model performance across various configurations through multiple evaluation metrics. Section “[Discussion](#)” provides a comprehensive analysis of the results, highlighting their theoretical significance and managerial relevance. Finally, section “[Conclusion](#)” wraps up the study by outlining its main contributions, summarizing the key findings, and proposing avenues for future research.

Literature review

In recent years, the rapid advancement of AI has significantly expanded its applications in the field of financial forecasting, notably in stock price prediction¹⁷. Concurrently, the widespread use of social media platforms has provided valuable opportunities to analyze investor sentiment, which has developed as a critical factor impacting market behavior and improving the performance of predictive models. The integration of AI-driven analytical techniques with SA enables researchers and investors to capture both quantitative trends and qualitative market perceptions, thereby enhancing predictive precision and decision-making effectiveness. Accordingly, identifying and examining these interconnected domains—AI-based forecasting, SA, and multimodal data integration—plays a crucial role in developing more robust and precise predictive models. The following subsections review the existing literature across these three areas to establish the theoretical foundation for the present study.

Stock price prediction using AI algorithms

Recent work consistently shows a shift from classical predictors (e.g., ARIMA or linear models) toward ML and DL, chiefly because market dynamics are nonlinear, high-dimensional, and temporally dependent. A representative review reports renewed interest in recurrent architectures for capturing temporal structure in

financial time series and notes their empirical advantages over traditional baselines, specifically when price sequences exhibit long-range dependencies^{18,19}. Complementing this perspective, Wang et al.²⁰ present an LSTM-based model aimed at overcoming the shortcomings of traditional RNNs in stock prediction. By using gating mechanisms to manage long-term dependencies and applying preprocessing, denoising, and regularization, their approach enhances stability and improves forecasting accuracy, showing LSTM's effectiveness for financial time-series modeling. Meanwhile, Mu et al.²¹ propose a model that integrates investor sentiment with historical trading data to enhance stock price prediction. The study builds a stock-specific sentiment dictionary and uses the Sparrow Search Algorithm to optimize LSTM hyperparameters. Results show that combining sentiment indicators with optimized deep learning improves forecasting robustness and captures market behavior more effectively.

Empirically, Li et al.²² develop an LSTM-based model to predict the closing prices of major technology stocks using a decade of historical data. Their two-layer LSTM architecture captures temporal dependencies in price movements and demonstrates reliable trend-learning capability. The study highlights LSTM's strength in modeling nonlinear financial time series and its suitability for stock market forecasting. At the application level, Varadharajan et al.²³ investigate an LSTM-RNN framework for predicting Amazon's closing stock prices. By comparing hyperparameter settings and validating performance on both synthetic ARIMA-based data and real market series, the study shows that LSTM networks effectively learn nonlinear temporal patterns and closely track actual price movements, confirming their suitability for robust financial time-series forecasting.

Another line of evidence shows that prediction improves when LSTMs are combined with information beyond raw prices. Vargas et al.²⁴ introduce a hybrid deep-learning framework that combines financial news titles with technical indicators for predicting daily stock movements. Using a CNN to extract textual semantics and an LSTM to model temporal trends, their SI-RCNN model outperforms indicator-only baselines and shows the strong predictive value of news-driven features. Convolutional architectures also matter especially for volatility and feature-rich inputs. Liu²⁵ proposed an enhanced CNN model for stock-volatility forecasting, showing that convolutional networks effectively capture both local and global structures in noisy, high-dimensional financial data. By modeling volatility-driven patterns, the approach outperforms traditional ML methods and even surpasses some sequence-based models in volatility-focused settings.

Alkhatib et al.²⁶ introduce an active deep-learning framework for stock price forecasting that evaluates six neural architectures, including MLP, GRU, LSTM, Bi-LSTM, CNN, and CNN-LSTM, across multiple companies. By engineering an expanded feature set, the study demonstrates that incorporating additional price-derived variables can enhance predictive stability and improve overall model performance. Lei et al.²⁷ introduced the TFI-DRL framework, combining deep learning and reinforcement learning with feature-gating and temporal-attention modules. The model effectively selects key indicators and captures time patterns, outperforming benchmark trading strategies in profit, return, and Sharpe ratio. Bansal et al.²⁸ compared five ML models KNN, LR, SVR, DTR, and LSTM for predicting stock prices of 12 Indian companies using seven years of data. Results showed that LSTM achieved the highest accuracy due to its strength in modeling nonlinear and time-dependent patterns, highlighting the superiority of DL over traditional ML methods.

Also, Recent studies have also explored network-based and information-theoretic approaches for financial prediction. Choi and Kim²⁹ proposed ETF forecasting frameworks that use statistical dependence-based and information-theoretic networks, including measures such as mutual information and transfer entropy, along with node embeddings, centrality measures, ensemble learning, and explainable AI techniques such as SHAP. Additionally, Choi and Kim³⁰ on information-theoretic networks and node embeddings for ETF prediction demonstrates how centrality measures derived from financial networks can serve as informative features for gradient-boosting models. Their finding that network topology encodes predictive information about price dynamics is directly relevant to the current study's claim of capturing "market dynamics."

Similarly, Kim et al.³¹ showed that effective transfer entropy can serve as a useful market explanatory variable and improve prediction performance across models such as LSTM and XGBoost. These studies demonstrate that network topology and inter-asset dependency structures can provide valuable predictive information. Chen and Wei³² emphasized the importance of incorporating information from companies related to a target firm when predicting stock prices. They constructed a company graph based on real-market investment relationships and applied node-embedding techniques to generate distributed representations for each firm. Their results showed that these graph-based representations can effectively capture inter-company relationships and improve the accuracy of stock price prediction models. In contrast, the present study focuses on single-asset forecasting through temporal multimodal fusion of historical prices, technical indicators, and FinGPT-derived sentiment features, rather than explicitly modeling cross-asset relational networks.

Another notable point in the existing literature is the growing attention paid to the use of multi-source data and hybrid frameworks in financial modeling and forecasting. Nti et al.³³ proposed a multi-source information-fusion framework for stock price prediction using a hybrid deep learning architecture based on CNN and LSTM, named IKN-ConvLSTM. They integrated stock-related information from six heterogeneous sources, including numerical and textual data, to address the challenge of combining diverse financial indicators. Their empirical results on Ghana Stock Exchange data showed that fusing multiple data sources significantly improved prediction accuracy compared with using individual datasets separately. Ke et al.³⁴ proposed a multi-source deep learning framework to predict cryptocurrency pin-bar reversal risk events by integrating blockchain metrics, social sentiment, and regulatory signals. Using an LSTM-based architecture, they showed that temporal anomalies can be effectively captured, while blockchain-related variables made a substantial contribution to predictive performance. Their findings also indicated that combining diverse data sources improves rare-event detection and offers practical value for financial risk monitoring under uncertain market conditions.

Ke et al.³⁵ developed a hybrid methodological framework for technical feature analysis using high-frequency financial data. Their approach combined CNN for extracting local price-volume features, GRU for capturing

temporal dependencies, and an improved genetic algorithm for dynamic hyperparameter optimization. They also introduced a multi-time-window fusion strategy and Bayesian adaptive weighting to integrate deep learning factors across different horizons. Their empirical findings showed that the proposed hybrid factor achieved stable performance and outperformed individual factors in both Chinese and U.S. markets. Accordingly, and in line with this growing research trend toward multi-source data integration and hybrid modeling frameworks, the present study proposes a new CNN-Attention-based LSTM model that combines price data, technical indicators, and Twitter sentiment to enhance financial prediction performance.

Technical Indicators in stock price prediction

The integration of TIs into stock price prediction models has become a cornerstone of contemporary financial forecasting, offering structured, quantitative insights into market behavior. Numerous studies emphasize that when used in conjunction with ML techniques, TIs substantially enhance the accuracy, interpretability, and robustness of prediction systems. Several studies adopt traditional and advanced ML algorithms to explore the predictive power of TIs. For instance, Ayala et al.³⁶ combine MACD and TEMA with ML models such as SVR, ANN, and Random Forests. Their findings show that hybrid models, which blend technical strategies with ML techniques, outperform purely statistical or indicator-only approaches in generating actionable trading signals. Similarly, Saetia and Yokrattanasak³⁷ incorporate indicators like RSI, SMA, WMA, and MACD alongside Google Trends data and ML algorithms such as XGBoost and Random Forest. Their study highlights that combining structured indicator data with auxiliary variables (like search trends) can lead to higher predictive performance under both normal and crisis conditions.

DL architectures also benefit from the inclusion of technical indicators. Fathali et al.³⁸ evaluate LSTM, RNN, and CNN models on the Indian NIFTY 50 index and demonstrate that LSTM when coupled with effective feature selection based on TIs yields the lowest prediction MSE, reinforcing the idea that carefully selected indicators can guide deep temporal models to learn more meaningful patterns. A similar conclusion is drawn by Gao et al.³⁹, who integrate multiple TIs into both LSTM and GRU architectures. Using LASSO and PCA for dimensionality reduction, they show that LASSO-based TI selection enhances model accuracy more than PCA, and both recurrent models perform well, though neither dominates decisively. The development of hybrid frameworks further underlines the role of TIs in predictive modeling.

Albahli et al.⁴⁰ propose new model, which uses an autoencoder to reduce correlation among 18 technical indicators and feed the optimized representation into a 1D DenseNet. Their system not only predicts stock prices but also outputs trading decisions (buy/sell/hold), achieving state-of-the-art results in terms of MAPE. The paper underscores the dual role of TIs in both feature engineering and downstream classification in deep models. Moreover, integrating multimodal data sources including TIs has shown to boost model robustness. Liu⁹ proposes a multimodal factor-mining approach that fuses market data, financial text, social sentiment, and technical indicators using GNNs and NLP methods. This integration captures hidden cross-modal patterns and shows that technical indicators work synergistically with sentiment-based inputs to improve model robustness.

Mostafavi and Hooman⁴¹ examined 88 technical indicators across momentum, trend, volatility, and volume groups to predict S&P 500 prices using RE, XGBoost, SVR, and LSTM with PCA. LSTM achieved the best accuracy, while TEMA, Ichimoku, and MFI were the most influential features, offering a data-driven framework for improving stock forecasting. Finally, Vargas et al.²⁴ show that combining technical indicators with financial news enhances forecasting accuracy. Their hybrid CNN-LSTM model learns from both numerical TI patterns and textual sentiment, outperforming models that rely on only one type of data.

Sentiment analysis

A first stream of evidence shows that text-derived sentiment provides incremental signal beyond prices alone. Nguyen et al.⁴² used message-board discussions to extract topic-specific sentiment signals and showed that incorporating these sentiment features into a price-only model noticeably improves directional prediction accuracy especially for stocks that are typically more difficult to forecast. Their findings indicate that fine-grained, topic-conditioned sentiment helps explain market variations that technical features alone fail to capture. In a complementary framework that integrates news sentiment with historical price data, Khedr and Yaseen⁴³ demonstrate that this fusion leads to substantially improved predictive accuracy compared to relying on either data source alone. Their results highlight how sentiment information helps models better distinguish market regimes and reduce noise during volatile periods.

Koukaras et al.⁴⁴ show that social-media microstructure from Twitter and StockTwits becomes highly predictive when paired with modern ML models. Combining microblog sentiment extracted by VADER with classical classifiers especially SVM significantly improves stock-movement prediction, indicating that concise polarity signals from retail investor posts meaningfully enhance model performance. Beyond Twitter, Bharathi & Geetha⁴⁵ exploit RSS news feeds and market indices; fusing these sources delivers accuracy gain over conventional baselines, indicating that SAs benefits generalize across text channels and not merely social platforms.

DL studies further argue that sentiment is especially potent when fused with sequence models that capture temporal dependence. John & Latha¹⁰ introduce a hybrid recurrent architecture (stacked LSTM/GRU with Bi-LSTM) and show the model's performance improves when financial-news sentiment is concatenated with price features, outperforming single-family RNNs and traditional statistics consistent with sentiment providing higher-level context that temporal encoders can retain and exploit. Pushing this idea, Jun Gu et al.⁴ couple FinBERT (domain-specific transformer for financial text) with an LSTM over market data: FinBERT-LSTM surpasses vanilla LSTM and DNN on MAE/MAPE/accuracy, illustrating that finance-tuned language models extract richer tonal and event signals than generic lexicons, which then translate into tighter forecasts when synchronized with price histories.

Under high-uncertainty market conditions, SA appears to improve robustness as well as accuracy. A study on U.S. and European bank stocks shows that combining LSTM forecasts with FinBERT-based sentiment from tweets and news outperforms models that depend only on long historical price data. The results indicate that sentiment analysis captures rapid narrative shifts that price-only models fail to detect during sudden market regime changes⁴⁶. Awad et al.⁴⁷ propose a DRL framework that fuses LSTM, ANN, NLP, and sentiment analysis using historical prices, news, and tweets. Their model combines BERT-based sentiment features, VMD-LSTM temporal learning, and Deep Q-learning for trading decisions. Results show this hybrid, sentiment-aware approach improves forecasting accuracy and trading performance over traditional methods.

Sousa et al.⁴⁸ evaluated BERT for sentiment analysis of financial news to enhance stock prediction. After fine-tuning BERT on a labeled news dataset, the model surpassed classical methods by capturing richer contextual meaning. Their analysis showed that extracted sentiment trends align closely with major market movements, indicating that transformer-based models offer strong potential for real-time sentiment tracking and short-term financial forecasting. Pagolu et al.⁶ examined the link between Twitter sentiment and stock market movements using large-scale Microsoft-related tweets. By representing text with Word2Vec and N-grams and classifying sentiment via Random Forest, they showed that aggregated online sentiment over short windows strongly aligns with upcoming price shifts. Their findings highlight the value of social-media emotions for anticipating investor behavior and improving short-term market forecasts.

Peivandizadeh et al.⁴⁹ introduced a TLSTM model combined with social-media SA to enhance stock forecasting. Their approach tackles imbalanced sentiment data and temporal dependencies by using an off-policy PPO classifier alongside TLSTM for dynamic pattern learning. By integrating sentiment signals from news and Twitter with historical prices, the framework captures both behavioral and market dynamics, outperforming traditional LSTM- and sentiment-only methods.

Table 1 summarizes the main characteristics of the studies reviewed in this research, including their prediction models, data sources, SA methods, and case study settings. This consolidated overview highlights existing methodological trends and differences across prior work, while also helping to identify gaps that motivate the development of the proposed framework.

Research gap

The present study introduces several methodological and conceptual innovations intended to improve the predictive accuracy and interpretability of stock price forecasting models. In contrast to conventional methods that depend only on numerical or technical information, this research integrates behavioral sentiment signals with temporal and spatial learning architectures. The following contributions summarize the key advancements of this study in terms of data integration, model architecture, and empirical validation.

- 1. Integration of three complementary data sources:** This study integrates three heterogeneous but complementary data types: historical prices, technical indicators, and textual sentiment from Twitter into a unified forecasting framework. Historical data capture long-term market dynamics through variables such as opening, closing, high, low, and volume. Technical indicators (e.g., RSI, MACD, MA/EMA, and Williams %R) enhance these signals by quantifying momentum, volatility, and short-term trends, while textual sentiment introduces behavioral information derived from investor opinions. The combination of these rational (quantitative) and behavioral (qualitative) features allows the model to better represent market dynamics, improving prediction stability and reducing forecasting errors compared to single-source models.
- 2. Sentiment quantification using FinGPT and polarity scoring:** Unlike traditional lexicon-based or static NLP approaches, this research employs FinGPT, an advanced financial language model capable of context-aware sentiment interpretation. For each tweet, FinGPT outputs probabilities of positive, negative, and neutral sentiment. A polarity score is then computed as, ranging from -1 to $+1$. The daily average polarity score for each stock is calculated by aggregating all tweet scores within the same day. This continuous numeric feature allows unstructured textual sentiment to be transformed into a structured input for time-series models. FinGPT's dynamic language understanding minimizes misinterpretation caused by financial jargon or sarcasm, producing more reliable sentiment signals for forecasting models.
- 3. Development of a novel hybrid ALSTM–CNN model with attention mechanism:** A new hybrid DL architecture is proposed that combines ALSTM and CNN components. The LSTM captures long-term temporal dependencies, the CNN extracts short-term and local features from financial time series, and the attention mechanism assigns adaptive importance weights to influential time steps. This hybridization enhances both the representational power and interpretability of the model. Compared to single-architecture approaches, the hybrid model can simultaneously learn slow-moving patterns and sudden fluctuations, yielding higher accuracy and robustness in dynamic market conditions.
- 4. Fusion of sentiment features with historical and technical indicators:** The DPS are fused with historical and technical features to create a comprehensive input matrix (10 features for the sentiment-based scenario). This integration enables the model to incorporate investor mood and market psychology alongside traditional numerical indicators. Empirical results demonstrate that adding sentiment features significantly improves prediction accuracy.
- 5. Real-world case study on eight major stocks:** The proposed framework was evaluated on eight major U.S. stocks: AMD, AMZN, AAPL, GOOG, MSFT, PG, TSLA, and TSM over a one-year period (September 2021–September 2022). This period selection ensured alignment between historical data (from Yahoo Finance) and textual data (from Kaggle/Twitter). The dataset diversity, spanning sectors from technology to consumer goods, allowed for robust cross-domain testing. The results confirmed that the hybrid model with sentiment integration consistently achieved superior accuracy across both volatile and stable stocks, demonstrating high generalization capability under real market conditions.

No.	Authors	Year	Prediction model	Features			Case study	Textual data source	SA model	Time frame	Interpret ability	Ref.
				Historical data	Textual data	Technical indicator						
1	Abolmakarem et al.	2023	LSTM, SVR, LR, RF, RR	Yes	No	Yes	Selected companies	No	No	Daily	No	50
2	Ma et al.	2023	AE, LSTM,	Yes	No	No	Selected companies	No	No	Daily	No	51
3	Colasanto et al.	2022	MLP, LSTM, RF, RNN, LOG	Yes	Yes	No	Selected companies	Financial times	BERT	Hourly	No	52
4	Mu et al.	2023	MS-SSA-LSTM	Yes	Yes	No	Selected companies	East Money Forum	Gooseeker	Daily	No	21
5	Rao & Srivastava	2012	ARIMA	Yes	Yes	No	NASDAQ, DJIA	Twitter	Naive Bayes	Daily, Weekly, Monthly	No	53
6	John & Latha	2023	HyRNN-Bi-LSTM	Yes	Yes	No	DJIA	NASDAQ website	VADER	Daily	No	10
7	Sousa et al.	2019	No	Yes	Yes	No	DJIA	Websites	BERT	Daily	No	48
8	Khedr & Yaseen	2017	KNN, SVM	Yes	Yes	No	Selected companies	Websites	Naive Bayes	Daily	No	43
9	Fatouros et al.	2023	No	Yes	Yes	No	FOREX	FOREX News	GPT	Daily	No	54
10	Mostafavi & Hooman	2025	LSTM, XGBoost, RF, SVR	Yes	No	Yes	S&P 500	No	No	Daily	No	41
11	Leow et al.	2021	No	Yes	Yes	No	ETFs	Twitter	BERT, VADER	Daily	No	55
12	Jun Gu et al.	2024	LSTM, DNN	Yes	Yes	Yes	NASDAQ	Websites	FinBERT	Daily	No	4
13	Bacco et al.	2024	LSTM	Yes	Yes	No	26 major banks	Twitter, Websites	FinBERT	Daily	No	46
14	Zhao and Yang	2023	LSTM, LS-RF, LR, SVM, SA-DLSTM	Yes	Yes	No	HIS market	Websites, social media	Sentence-based method	Daily	No	56
15	Wang et al.	2020	RNN-LSTM-CNN	Yes	No	Yes	Selected companies	No	No	Daily	No	57
16	Fathali et al.	2020	GRU	Yes	No	Yes	NIFTY 50 Index	No	No	Daily	No	38
17	Dezhkam & Manzuri	2023	HHT-XGB	Yes	No	No	S&P 500	No	No	Daily	No	58
18	Awad et al.	2023	DQN, LSTM, ANN	Yes	Yes	Yes	Gold	Social media	BERT, TFIDF	Daily	No	47
19	Behera et al.	2023	RE, ADB, XGB, SVM, KNN, ANN	Yes	No	No	Selected companies	No	No	Monthly	No	59
20	Abdelfattah et al.	2024	LSTM	Yes	Yes	No	Selected companies	Twitter	VADER	Daily	No	60
21	Taheripour et al.	2025	No	Yes	Yes	No	DJIA	Quarterly report	FinBERT	Seasonally	No	61
22	Bansal et al.	2022	LR, KNN, SVR, DTR, LSTM	Yes	No	No	Bombay Stock Exchange	No	No	Daily	No	28
23	Mehta et al.	2021	LSTM	Yes	Yes	Yes	NSE India	Social media, Websites	Naive Bayes	Daily	No	62
24	Mukherjee et al.	2023	FFNN, CNIN	Yes	No	No	NIFTY	No	No	Monthly	No	63
25	Peivandizadeh et al.	2024	TLSTM	Yes	Yes	No	NSE India	Social media	BERT	Daily	No	49
26	Akhtar et al.	2022	LSTM, RF, SVM	Yes	No	No	BSE500, DJIA, NASDAQ	No	No	Daily	No	13
27	Liu et al.	2022	DL	Yes	No	Yes	SSE China, SZSE China	No	No	Daily	No	64
28	Vargas et al.	2018	CNN-LSTM	Yes	Yes	Yes	NYSE	financial news	No	Daily	No	24
29	Carosia et al.	2025	MLP, LSTM, BiLSTM	Yes	Yes	Yes	Brazilian stock market	Newspaper	Word2Vec embeddings	Daily	No	65
30	Saravanos & Kanavos	2025	LR, SVR, RF, ADB, XGB, GB, LSTM, GRU, CNN, GAN	Yes	Yes	No	Selected companies	Twitter	N-grams	Daily	No	66
Our study		CNN, LSTM, ALSTM,Hybrid		Yes	Yes	Yes	Selected company	Twitter	FinGPT	Daily	SHAP	

Table 1. Overview of key studies reviewed, including models, data types, and SA approaches.

6. **Systematic experimental design and comparative evaluation:** A structured experimental setup was designed to evaluate four models CNN, LSTM, ALSTM, and Hybrid under two conditions: with and without SA. Data were partitioned into 80% for training and 20% for testing sets for all models to ensure fair comparison. This two-scenario framework quantitatively isolates the effect of sentiment inclusion, revealing that sentiment-enhanced models outperform baseline versions across all evaluation metrics. The hybrid model consistently achieved the best or near-best results, validating its robustness and adaptability in diverse stock environments.

Therefore, this research introduces an excellent hybrid technique for price prediction that simultaneously incorporates three distinct categories of input data: price-based, technical, and contextual information. The proposed architecture is designed with three main computational layers: LSTM, CNN, and an attention mechanism, each uniquely involved in modeling the complex temporal and spatial dependencies inherent in financial temporal data. The LSTM layer is employed to capture regular dependencies and long-term temporal relationships in historical price movements. The CNN layer focuses on extracting local and short-term patterns from the data, increasing the model's sensitivity to subtle market fluctuations. The AM then automatically adjusts the weights of features based on relevance, allowing the model to focus on influential information, thereby improving interpretability and prediction accuracy. In addition, contextual information is processed through the FinGPT model, which provides domain-specific sentiment representations. Together, these components form a robust multi-modal forecasting model capable of providing more accurate, interpretable, and stable stock price forecasts.

Case study and data collection

This section is organized into two main components. The first component provides a brief overview of the case study, explaining its background, significance, and reasons for considering it as a suitable context for the proposed model. The second component details the data set required for effective implementation of the model. In this section, specific data sets, their characteristics, and their relationship to the analytical framework are discussed in depth, providing a clear understanding of how each data type supports the operational structure of the model.

Case study

The case study in this research focuses on the merging of multi-source data to predict stock prices using historical, technical, and textual sentiment information. The textual data used in this study were obtained from a publicly available dataset on Kaggle, which contains tweets related to various stock tickers over a defined period from September 21, 2021, to September 21, 2022. To maintain consistency and ensure the temporal alignment of data sources, the same one-year period was selected for the collection of historical stock data from Yahoo Finance. The study investigates eight major publicly traded companies representing different sectors of the global market, each listed on the NASDAQ or NYSE. Each tweet was pre-processed by removing noise such as URLs, hashtags and special characters and then analyzed to extract sentiment features that indicate positive, negative or neutral opinions. These include AMD, a leading semiconductor manufacturer known for its CPUs and GPUs; AMZN, among the world's foremost companies in e-commerce and cloud computing; AAPL, a global technology firm specializing in consumer electronics and digital services; GOOG, the parent company of Google, dominating the internet services and digital advertising sectors; MSFT, a multinational technology company recognized for its software, hardware, and cloud solutions; PG, a multinational consumer goods corporation; TSLA, a prominent company in electric vehicles and renewable energy; and TSM, the world's largest semiconductor foundry. These firms were chosen to reflect a diverse range of sectors including technology, consumer goods, and manufacturing thereby enabling a comprehensive assessment of stock price prediction performance under varying market conditions.

The selection of the eight selected stocks was also guided by the availability of corresponding Twitter data in the Kaggle dataset. These companies were chosen because their social media activity provided sufficient sentiment information for reliable analysis. By aligning the temporal range of historical, technical, and textual data, this study ensured that all datasets were synchronized and comparable, allowing for an accurate and integrated evaluation of the relationship between market indicators, investor sentiment, and stock price dynamics. First, historical market data were gathered for each stock, including daily opening price, closing price, highest price, lowest price, and trading volume. These variables form the fundamental time series used for trend and volatility modeling. Figure 1 shows the price behavior of these companies in the form of a candlestick chart.

To provide a preliminary understanding of the stock price behavior during the observation period, Table 2 summarizes the main descriptive statistics for the eight selected stocks. The table presents the minimum, maximum, average and standard deviation of the daily closing prices for each stock. These statistics illustrate the volatility and overall performance range of each company's stock during the study period.

As shown in Table 2, the eight selected tickers exhibit substantial variation in their price ranges and volatility levels, reflecting differing market behaviors and risk profiles across companies. AMD, AMZN, and TSLA display relatively wide gaps between their minimum and maximum values, accompanied by higher standard deviations, indicating more pronounced fluctuations over the observed period. In contrast, companies such as PG and AAPL show narrower price ranges and lower levels of variability, suggesting more stable performance. MSFT and GOOG fall in an intermediate category, with moderate volatility and consistent average prices. Overall, the data illustrate a spectrum of market dynamics, from highly volatile growth-oriented firms to more stable, predictable stocks.



Fig. 1. Candlestick charts of the eight selected companies illustrating daily price movements and volatility.

Data collection

In this study, the data collection process is organized into two primary categories: textual data and historical data. The textual dataset is obtained from Twitter, providing real-time sentiment and qualitative signals that reflect public opinions and market-related discussions. The historical data, on the other hand, are divided into two subgroups: price data and a set of technical indicators derived from market movements. While the textual data offer contextual information that complements numerical trends, the historical dataset supplies the quantitative foundation required for modeling price behavior and detecting underlying patterns. By integrating

No.	Ticker	Min	Max	Mean	SD
1	AMD	66.300	161.910	108.0043	23.983
2	AMZN	102.310	184.803	142.3818	22.962
3	AAPL	128.174	178.880	157.5998	11.796
4	GOOG	97.148	149.297	125.584	14.714
5	MSFT	231.939	334.436	282.4367	27.729
6	PG	123.500	152.472	139.7433	7.167
7	TSLA	209.387	409.970	299.9881	45.865
8	TSM	69.374	133.177	96.42062	15.597

Table 2. Summary statistics of daily closing prices for the selected stocks.

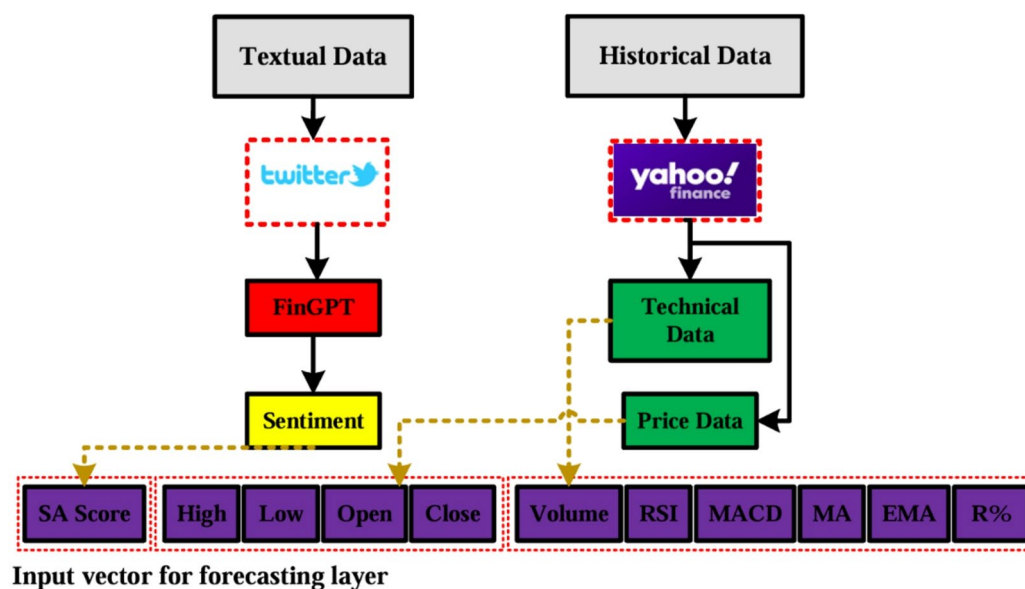


Fig. 2. The process of data gathering.

these different types of information, the study builds a more comprehensive and robust analytical framework for understanding market dynamics. Figure 2 illustrates the overall data collection procedure.

We will examine each of these categories below.

Price data

The historical stock data used in this study were collected over a one-year period for the eight selected companies. The data were obtained from the Yahoo Finance database, which provides publicly available and reliable financial market information. The dataset includes daily trading records consisting of the following key attributes:

- **Open price:** the stock price at the beginning of the trading session for a given day.
- **Close price:** the final price of the stock at the end of the trading day, often used as a reference point for performance evaluation.
- **High price:** The highest price at which the stock was exchanged throughout the trading day.
- **Low price:** The lowest price at which the stock was exchanged throughout the trading day.
- **Volume:** The overall volume of shares exchanged during the trading day, indicating the level of market activity and investor interest.

These features collectively describe the daily behavior and volatility of each stock and serve as the primary inputs for calculating various technical indicators and modeling price movements in the subsequent stages of analysis. A general overview of the stock price movements during the study period is demonstrated in Fig. 2, which presents the candlestick charts for the eight selected stocks. Every candlestick denotes the market activity for one trading day and visually summarizes the daily price behavior through four key values: opening price, closing price, highest price, and lowest price. Each candlestick's body shows the variation between the opening and closing values, while the wicks (or shadows) depict the intraday volatility. Green candlesticks indicate trading days in which the closing price exceeded the opening price. (price increase), and red candlesticks indicate the

opposite (price decrease). This visualization provides an intuitive understanding of market trends, fluctuations, and potential turning points over the one-year observation period.

Technical indicators

Technical indicators are essential in improving the predictive accuracy of stock price prediction models by capturing hidden market patterns, trends, and momentum that are not directly observable in raw historical data. Incorporating such indicators helps models better represent investor behavior and market dynamics, leading to improved predictive performance. For instance, Patel et al.⁶⁷ demonstrated that incorporating technical indicators like RSI, MACD, and MA into ML models markedly enhances forecasting accuracy compared with models relying solely on price data. In this study, to further improve prediction performance, five technical indicators were calculated for each trading day based on the closing price of the stocks. These indicators were incorporated as additional features in the input of the predictive models to enhance their learning capability and accuracy. The five selected indicators and their computational principles are briefly described in the following subsections.

RSI:

In technical analysis, the RSI ranks as one of the most frequently utilized momentum indicators, introduced by J. Welles Wilder⁶⁸. RSI calculates the magnitude and velocity of recent price changes to assess whether a stock is in an overbought or oversold condition. This indicator facilitates the identification of potential turning points and evaluates market momentum by analyzing the relative magnitude of recent advances and declines over a typical 14-day window.

$$RSI = 100 - \frac{100}{1 + RS} \quad (1)$$

$$RS = \frac{\text{Average Gain}}{\text{Average Loss}} \quad (2)$$

In this formulation, Average Gain represents the mean of all upward price movements, over the selected period, and Average Loss represents the mean of downward price movements (in absolute terms). RSI values range between 0 and 100, where values above 70 typically indicate overbought conditions (possible downward correction), and values below 30 suggest oversold conditions (potential upward movement).

The RSI is particularly useful in stock price forecasting because it quantifies market momentum and can reveal hidden divergences between price movements and relative strength, which are valuable for predictive modeling in both traditional and ML frameworks⁶⁹.

MA:

In technical analysis, the MA is considered one of the most essential and commonly utilized indicators. It is primarily employed to reduce the impact of short-term variations in price data and to highlight the fundamental trend direction of a financial asset over a specified period. By averaging past prices, the MA reduces noise and allows analysts and predictive models to better capture long-term market tendencies.

Mathematically, the SMA over a period n is defined as:

$$MA_t = \frac{1}{n-1} \sum_{i=0}^{n-1} p_{t-i} \quad (3)$$

where MA_t denotes the MA value at time t , p_{t-i} represents the closing price i days before t , and n is the chosen time window (commonly 10, 20, or 50 days). The MA acts as a trend-following indicator, signaling potential buy or sell opportunities when the actual price moves over or under the MA line. For instance, an upward crossover (price > MA) may indicate bullish momentum, while a downward crossover (price < MA) can signal bearish sentiment.

In predictive modeling, incorporating MA as an input feature helps capture trend information that enhances model stability and forecasting performance. Previous research has confirmed that including MA-based features improves the predictive accuracy of ML and DL models⁷⁰.

MACD:

The MACD is a momentum-based, trend-following indicator created by Gerald Appel⁷¹. The indicator is constructed to show how two MA of a stock's price interact, thereby assisting in identifying momentum changes, trend direction, and potential buy or sell signals. The MACD captures both trend strength and momentum shifts, making it among the most commonly used indicators in both traditional technical analysis and ML based stock prediction studies.

The formula is:

$$\text{MACD Line} = \text{EMA}_{12} - \text{EMA}_{26} \quad (4)$$

In predictive analytics, the MACD provides valuable information about trend reversals and momentum acceleration, serving as an effective input feature for forecasting models. Empirical studies have demonstrated that the integration of the MACD indicator leads to improved predictive accuracy in ML driven financial forecasting models^{69,72}.

EMA:

The EMA is an advanced form of the SMA that places greater emphasis on the most recent prices, thereby enhancing its sensitivity to ongoing market dynamics⁷³. Unlike the SMA, which treats all observations equally,

the EMA highlights the most recent price movements, thereby capturing short-term market dynamics more effectively. It is extensively used in technical analysis and predictive modeling to detect trend reversals and short-term momentum shifts.

$$EMA_t = \alpha \times P_t + (1 - \alpha) \times EMA_{t-1} \quad (5)$$

Where:

- P_t : is the closing price at time t
- EMA_{t-1} : is the EMA value of the prior day
- α : is the smoothing factor, calculated as:

$$\alpha = \frac{2}{1 + n}$$

here, n represents the number of periods used for the MA (e.g., 10, 12, 26). The EMA gives more significance to recent prices, enabling the indicator to react faster to new information compared to the SMA.

In forecasting applications, the EMA is often employed as an input feature to capture the smoothed representation of price trends, helping ML and DL models to reduce noise and enhance predictive accuracy. Previous studies have shown that EMA-based features contribute to improved performance in stock price prediction models^{70,74}.

Williams %R:

The Williams %R is a momentum oscillator developed by Larry R. Williams⁷⁵. It measures the level of the closing price related to the highest high and lowest low over a specific look-back period, typically 14 days. The indicator helps identify overbought and oversold market conditions and is particularly useful for detecting potential price reversals.

$$R\% = \frac{(Highest\ High - Lowest\ Low)}{(Highest\ High - Close)} \times (-100) \quad (6)$$

where:

- Highest High is the maximum price during the selected period,
- Lowest Low is the minimum price during the same period,
- Close is the most recent closing price.

The resulting values range from 0 to -100, where readings between 0 and -20 typically indicate overbought conditions, while values between -80 and -100 suggest oversold conditions. Unlike the RSI, which scales from 0 to 100, Williams %R is inverted, providing a complementary perspective on price momentum.

In stock price prediction, Williams %R captures short-term market sentiment and momentum shifts, making it a valuable input feature for ML models. Empirical research has shown that the inclusion of oscillators such as RSI and Williams %R can significantly improve the performance of forecasting models by providing nonlinear information about market reversals^{69,72}.

Textual data

To incorporate market sentiment into the predictive framework, textual data were collected from Twitter, focusing on tweets related to the eight selected stocks. The tweets were collected using the SnsCrape tool, which enables efficient extraction of publicly available Twitter data based on keyword and time filters. In this research, a dataset containing 69,290 tweets related to various stock tickers was obtained from Kaggle. The dataset is publicly accessible, and the corresponding data availability link is provided in the Data Availability Statement section of this paper.

An example of the collected Twitter data related to each of the eight selected stocks is shown in Table 3. The table reports the number of tweets gathered for each company during the observation period, along with a sample tweet representative of the content typically associated with each stock. This overview provides an insight into the distribution and nature of textual data used for SA in this study.

For SA of the gathered textual data, this research utilizes FinGPT, a finance-oriented language model specifically trained to interpret and analyze financial language. FinGPT is an open-source LLM framework specifically adapted for financial NLP tasks, such as SA of news articles, tweets, and other market-related texts. Through instruction-tuning and domain-specific fine-tuning (e.g., using Low-rank Adaptation, LoRA), FinGPT enables the model to map financial texts into structured sentiment categories positive, neutral, or negative and produce robust polarity scores. In Kang and Choi⁷⁶ study, SA was conducted on categorized news data using GPT-4o and FinBERT. In that research, news data were obtained both from Kaggle and through dynamic collection from The New York Times, and the models were evaluated based on headlines, descriptions, and different news sections. The results showed that GPT-4o, with optimized prompt design, outperformed the fine-tuned FinBERT model across all sections. These findings suggest that GPT/FinGPT-based approaches may achieve superior performance over FinBERT in financial sentiment analysis.

Unlike general-purpose LLMs, FinGPT addresses the distinctive challenges of financial language (such as numeric references, ticker symbols, domain jargon and semantic nuance) by leveraging curated data sources and financial-specific tasks. For instance, one study achieved higher accuracy than standard models through financial-instruction-tuning of FinGPT⁷⁷.

No.	Ticker	Number of Tweet	Example of Tweets
1	TSLA	37,422	@TeslaSolar roof stood up to #HurricaneIan with 155 mph winds and a 10 storm surge! This Powerwall was underwater for hours and is still working perfectly.@elonmusk @Tesla #powerwall #solarroof https://t.co/fVHmChyXUm
2	MSFT	4089	Scintriguing levels to add to (several I m already overweight):
3	PG	4088	Today's drop in \$SPX is a perfect example of what happens when \$AAPL and \$AMZN turn lower. There are a few names that have held up markets for years. Those names are finally dropping and we are beginning to see the damage. No healthy market is driven by just a few names.
4	AMZN	4088	Amazon, \$AMZN, unveils device that tracks how you breathe while sleeping
5	GOOG	1290	That's right everyone - \$GOOG is officially a shitco. 1 year ago it was the greatest business in existence. Now it is simply a second rate ad platform on top of aged out tech
6	AMD	2226	\$AMD - (Weekly) Update - Finally filling the imbalance today beautifully Made the callout 66P 0.80 currently 3.05 (281%) Even told the wolves its hitting 62 today This is what trading with 9+ years of experience looks like https://t.co/nWnjPCv1wWh https://t.co/cOJnTF22NU
7	AAPL	5055	\$AAPL was down almost 5% today. Rare to see it sell off like this. Headline was fears of demand slowly triggered the BofA downgrade. Despite the fears, demand seems to be favorable for iPhone 14. At least for today
8	TSM	11,032	I keep looking at the current #R6NAL standings and I still can't believe what I'm seeing. Anything Can Happen, right? (1) @SoniqsEsports. (2) @TSM. (3) @XSET. (4) @MirageSportE. (5) @DarkZeroGG. 6)@Spacestation. 7) @PBLMesports. 8) @Astralisgg. 9)@OXG_Esports. 10) @beastcoast https://t.co/MK6Zprc7GH

Table 3. Summary of collected Twitter data for each stock.

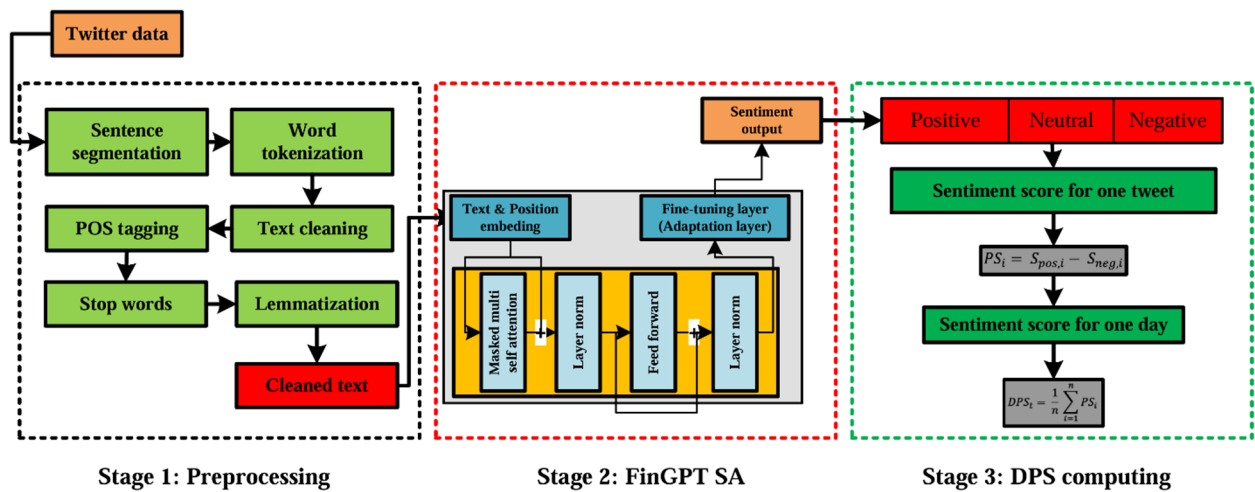


Fig. 3. FinGPT-based SA process.

The primary advantages of employing FinGPT in this research are as follows:

- *Domain awareness:* FinGPT is fine-tuned for financial text, improving accuracy in understanding investor sentiment, jargon, and market-specific expressions surpassing general-purpose LLMs in this context.
- *Accessibility and transparency:* As an open-source model, FinGPT allows full access to its architecture, training pipeline, and fine-tuning methods promoting reproducibility and cost-effective customization.
- *Seamless integration:* The output polarity score is directly compatible with numerical models, facilitating smooth fusion of sentiment with historical and technical features.

In summary, FinGPT serves as a robust tool to quantify investor sentiment at scale and transform it into actionable features for time-series forecasting models. Its inclusion in this study enhances the behavioral dimension of stock-price prediction and underpins the proposed hybrid architecture’s ability to capture market psychology alongside price dynamics.

Figure 3 illustrates the FinGPT-based SA framework, which is structured into three main stages: Preprocessing, FinGPT SA, and DPS Computation. Each stage plays a critical role in transforming raw textual data from social media into a meaningful numerical sentiment feature used for stock price forecasting.

In Stage 1 Preprocessing, raw Twitter data undergoes several NLP operations to ensure textual consistency and clarity. This stage includes sentence segmentation, word tokenization, text cleaning, POS tagging, removal of stop words, and lemmatization. These steps collectively remove irrelevant components such as punctuation and redundant words while standardizing terms into their base forms, resulting in a cleaned and structured dataset ready for analysis.

Stage 2 FinGPT SA involves processing the cleaned text through the FinGPT model, a fine-tuned large language model specifically trained for financial sentiment understanding. The model embeds both textual and positional information, applies masked multi-head self-attention mechanisms, and passes the representation

through feed-forward and normalization layers. A fine-tuning (adaptation) layer then outputs the sentiment classification for each tweet, categorized as positive, neutral, or negative.

In Stage 3 DPS Computing, sentiment scores are quantified numerically. For the SA process, this study employed the FinGPT model. FinGPT assigns three sentiment probabilities to each tweet positive, negative, and neutral based on its linguistic and contextual interpretation of the text. To quantify an overall sentiment tendency for each tweet, a PS was determined by subtracting the negative sentiment score from the positive one, defined as:

$$PS_i = S_{pos,i} - S_{neg,i} \quad (7)$$

In Eq. (7) $S_{pos,i}$, $S_{neg,i}$ represent the positive and negative sentiment probabilities for tweet i , respectively. The resulting polarity score ranges between -1 and $+1$, where positive scores indicate optimistic market sentiment and negative scores reflect pessimistic investor attitudes. For each stock, all tweets related to its ticker symbol (e.g., #TSLA, #AAPL) within a single day were aggregated, and the daily average polarity score was computed as:

$$DPS_t = \frac{1}{n} \sum_{i=1}^n PS_i \quad (8)$$

In Eq. (8) n indicates the total number of tweets about the stock on day t . This DPS was then used as a sentiment feature in the predictive modeling framework alongside historical and technical indicator data. Incorporating this variable enabled the models to account for investor sentiment dynamics, thereby improving their ability to capture behavioral aspects influencing stock price movements.

Methodology

In this section, we describe in detail the prediction models used in the research and fully explain the hybrid model developed to improve the prediction performance. These models include CNN, LSTM, ALSTM, and a hybrid model.

Convolutional neural network

The CNN, first developed by LeCun et al.⁷⁸, represents a DL architecture that transformed the field of pattern recognition. A standard CNN framework is composed of five fundamental components: an input layer, convolutional layers, pooling layers, fully connected layers, and an output layer. In the following section, the three principal components convolutional, pooling, and fully connected layers are discussed in detail. This Fig. 4 demonstrates the main architecture of the CNN model used in this study, demonstrating the hierarchical flow of data beginning at the input layer and moving through the convolutional and pooling stages before reaching the fully connected and output layers.

Convolutional layer: The convolutional layer extracts representative features from the input data using multiple learnable filters (kernels), each defined by its own set of weights and a bias term. In this layer, each neuron is connected only to a localized region of the preceding layer. In this layer, each neuron is linked exclusively to a specific local region of the previous layer, referred to as the receptive field, analogous to the receptive regions found in biological visual cortex cells. During the convolution process, the kernel slides over the input feature map, performing element-wise multiplications between the kernel and the covered portion of the input, summing the results, and then adding a bias value to produce the output feature map.

$$Y^{l+1}(c, d) = [Y^l \otimes w^{l+1}](c, d) + b = \sum_{K_l}^{f=1} \sum_f^{e=1} \sum_f^{y=1} [Y_l^k(s_0c + x, s_0d + y) w_{l+1}^k(e, f)] + b \quad (9)$$

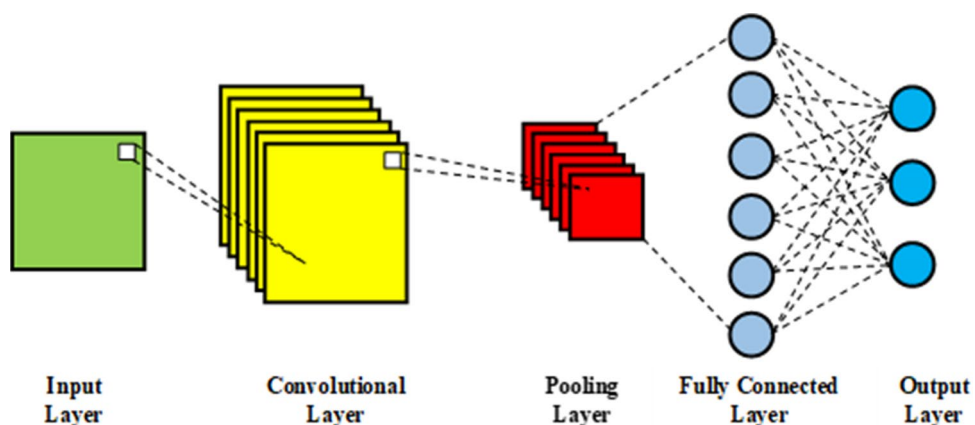


Fig. 4. The structural flow of the CNN model.

$$Y(c, d) \in \{0, 1, \dots, Z_{l+1}\}, Z_{l+1} = \frac{Z_l + 2q - f}{s_0} + 1 \quad (10)$$

The summation operation in Eq. (9) represents a cross-correlation process, where b denotes the bias term. $Y^{(l)}$ and $Y^{(l+1)}$ correspond to the input and output feature maps of the $(l+1)^{th}$ convolutional layer, respectively. Equation (10) defines $Y^{(l+1)}$ as the feature map's spatial dimensions, assuming uniform size across layers. Here, $Y(c, d)$ refers to the pixel intensity values, K indicates the number of channels, and s_0 , f , and q represent convolutional layer parameters. To enable the extraction of nonlinear and high-level representations, the convolutional layer applies an activation function, as described in Eq. (11).

$$A_{c,d,k}^l = f(Y_l^{c,d,k}) \quad (11)$$

A widely adopted activation function in convolutional networks is the ReLU, a nonlinear transformation that introduces sparsity and accelerates convergence during training. ReLU and its variants are mathematically formulated as ramp functions, as shown in Eq. (12).

$$f(e) = \max(0, e) \quad (12)$$

For an input vector e received from the preceding layer, the neuron applies a linear transformation $w^T e + b$, followed by a nonlinear activation function that outputs $\max(0, w^T e + b)$. This activated value is then propagated to the subsequent layer.

Pooling layer: Following feature derivation in the earlier layer, the resulting feature maps are passed to the pooling layer, which performs a filtering and dimensionality reduction process. The pooling operation substitutes each region with a representative statistical value such as the maximum or average computed over neighboring elements. This process is governed by parameters including pooling window size, stride, and padding, as expressed in Eq. (13).

$$f(e) = \max(0, e) \quad (13)$$

In Eq. (5), the step size s_0 and pixel coordinates (c, d) retain the same definitions as in the convolutional layer, while ss denotes a predefined parameter. The pooling operation calculates a representative value within each pooling region when $p = 1$, it performs average pooling, computing the mean of the local values; as $p \rightarrow \infty$, it becomes max pooling, choosing the highest value in the region. Both pooling methods are widely used in CNN architectures to reduce spatial dimensions while retaining essential texture and background information, though some fine details are inevitably lost in the process.

Fully connected layer: In CNN, the fully connected (FC) layer functions as the classifier that maps the high-level features derived by the convolutional and pooling layers onto the target output space. Rather than learning new spatial features, the FC layer integrates and combines the abstract representations learned in previous stages to generate the final prediction or class label.

Long short-term memory

The LSTM model, introduced by Hochreiter and Schmidhuber⁷⁹, was structured to effectively record sequential relationships within data. Although LSTMs belong to the broader family of RNNs, they surpass conventional RNNs by maintaining information over extended time intervals, thereby mitigating issues such as short-term memory loss and vanishing gradients⁷⁴.

The LSTM architecture includes an input layer, multiple hidden layers, and an output layer, where memory cells serve as the core components within the hidden layers. As illustrated in Fig. 5, each memory cell operates based on two key elements: x_t , representing the input at time step t , and h_t , denoting the corresponding hidden state. The three gates that regulate information flow the input gate (i_t), output gate (o_t), and forget gate (f_t) work together to manage how information is stored, updated, or removed within the network. The cell state (s_t) maintains long-term information by integrating new inputs and discarding irrelevant data. Specifically, the input gate determines what data enters the memory cell, the forget gate decides which data to erase, and the output gate controls what data is passed to the next time step. The mathematical formulations governing these gates and states are presented in the following Eq.

$$f_t = \text{sigmoid}(Q_{f,x}x_t + Q_{f,h}h_{t-1} + b_f) \quad (14)$$

$$\tilde{s}_t = \text{sigmoid}(Q_{\sim s,x}x_t + Q_{\sim s,h}h_{t-1} + b_{\sim s_t}) \quad (15)$$

$$i_t = \text{sigmoid}(Q_{i,x}x_t + Q_{i,h}h_{t-1} + b_i) \quad (16)$$

$$s_t = f_t * s_{t-1} + i_t * \tilde{s}_t \quad (17)$$

$$o_t = \text{sigmoid}(Q_{o,x}x_t + Q_{o,h}h_{t-1} + b_o) \quad (18)$$

$$h_t = o_t * \tanh(s_t) \quad (19)$$

In this context, the vector x_t corresponds to the existing entry. The parameters $Q_{f,x}$, $Q_{i,x}$, $Q_{\sim s,x}$, $Q_{o,x}$ are the weight matrices associated with the input vector for the forget, entry, candidate, and product gates, respectively.

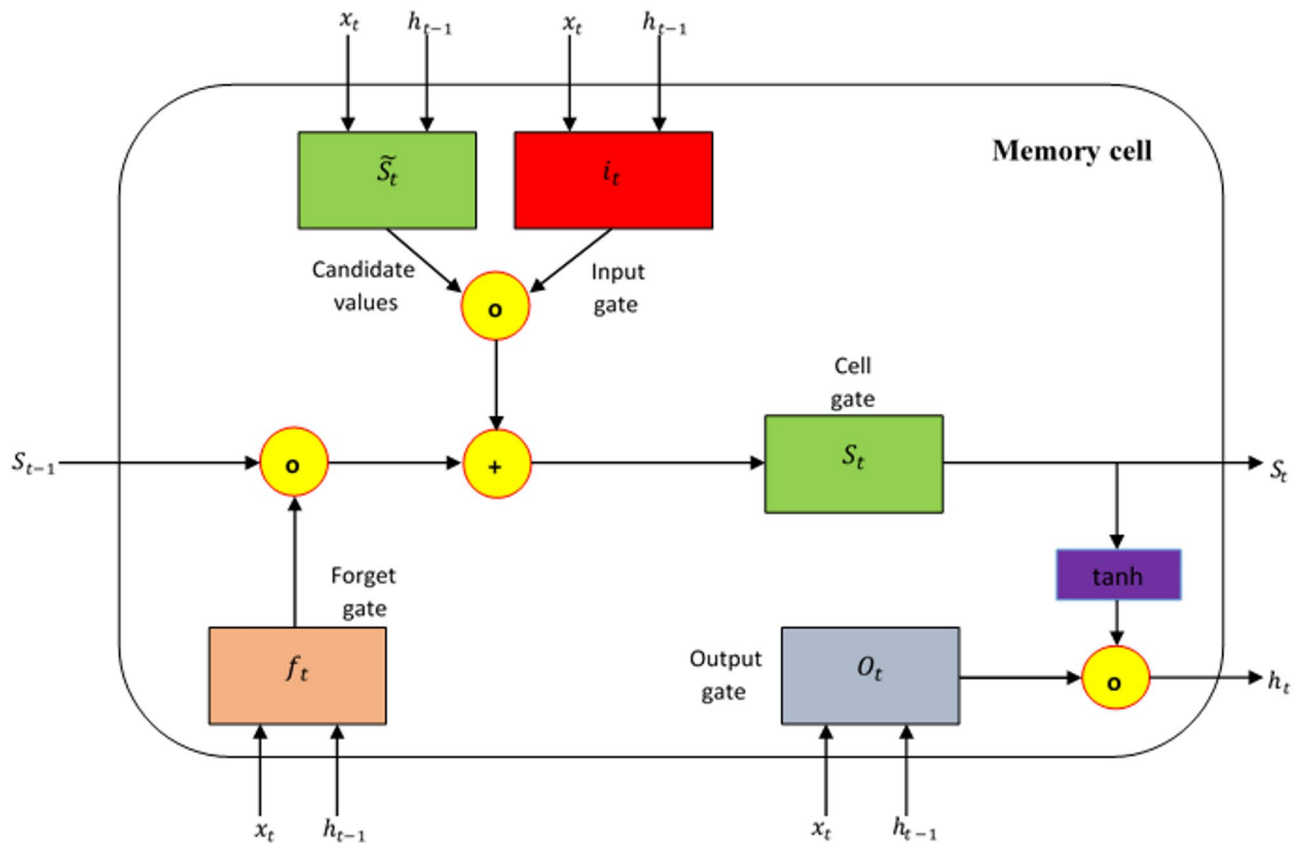


Fig. 5. The architecture of the LSTM memory cell.

The term h_{t-1} is the hidden state vector from the previous time step, with corresponding weight matrices $Q_{f,h}$, $Q_{i,h}$, $Q_{\tilde{s},h}$, $Q_{o,h}$. Finally, b_f , b_i , $b_{\tilde{s}}$ and b_o represent the bias vectors for each respective gate.

Attention-based LSTM

The ALSTM model represents an enhanced variant of the conventional LSTM, developed to overcome constraints in modeling long-term dependencies within time-series data. By integrating the memory retention capability of LSTM with an attention mechanism, the ALSTM can more effectively identify and emphasize important temporal patterns across extended sequences. This hybrid design enhances predictive accuracy in time-series forecasting tasks by simultaneously managing both short-term fluctuations and long-term trends. The model's robustness lies in its ability to selectively focus on the most relevant time steps, thereby improving interpretability and efficiency in sequential learning. Since the LSTM architecture has been described earlier, the following section provides a concise overview of the attention layer and its role in this framework.

Attention mechanism: The attention mechanism functions as a dynamic weighting process that assigns learned importance scores to different input values over time, effectively allowing the model to concentrate on the most informative parts of a series. Acting as an adaptive memory component, it enhances the model's ability to extract meaningful features and capture temporal dependencies more accurately. By prioritizing specific inputs, the attention mechanism mitigates information bottlenecks and improves learning efficiency. Moreover, it helps address the vanishing gradient issue by establishing direct connections among the encoder and decoder states, ensuring smoother information flow across time steps. Figure 6 illustrates the overall architecture, depicting how the mechanism receives input, computes attention scores, generates a probability distribution, and produces the final context-aware output.

The attention mechanism⁸⁰ operates through a sequential process involving the computation of attention scores, the assignment of weights, and the generation of context vectors. The alignment mechanism first calculates a score $e_{(t,i)}$ based on the encrypted hidden states h_i and the prior decoder output $S_{(i-1)}$, reflecting the degree of relevance between each input component and the current output. The attention score is obtained using a scoring function $a(\cdot)$, as expressed in Eq. (20). A softmax function is then used to these scores to normalize them into weights $\alpha_{(t,i)}$, as defined in Eq. (21). Subsequently, the content vector c_t is calculated as the weighted sum of the feature extractor hidden states and is passed to the decoder at each time step, as described in Eq. (22). This mechanism improves algorithmic effectiveness by decreasing memory needs and filtering out redundant or noisy information, enabling the model to focus discriminately on the most informative elements of the input series.

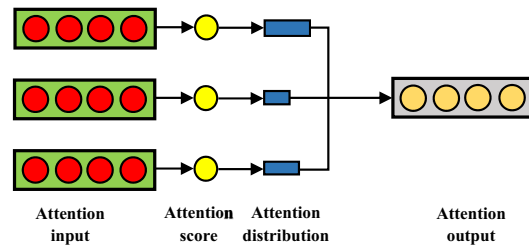
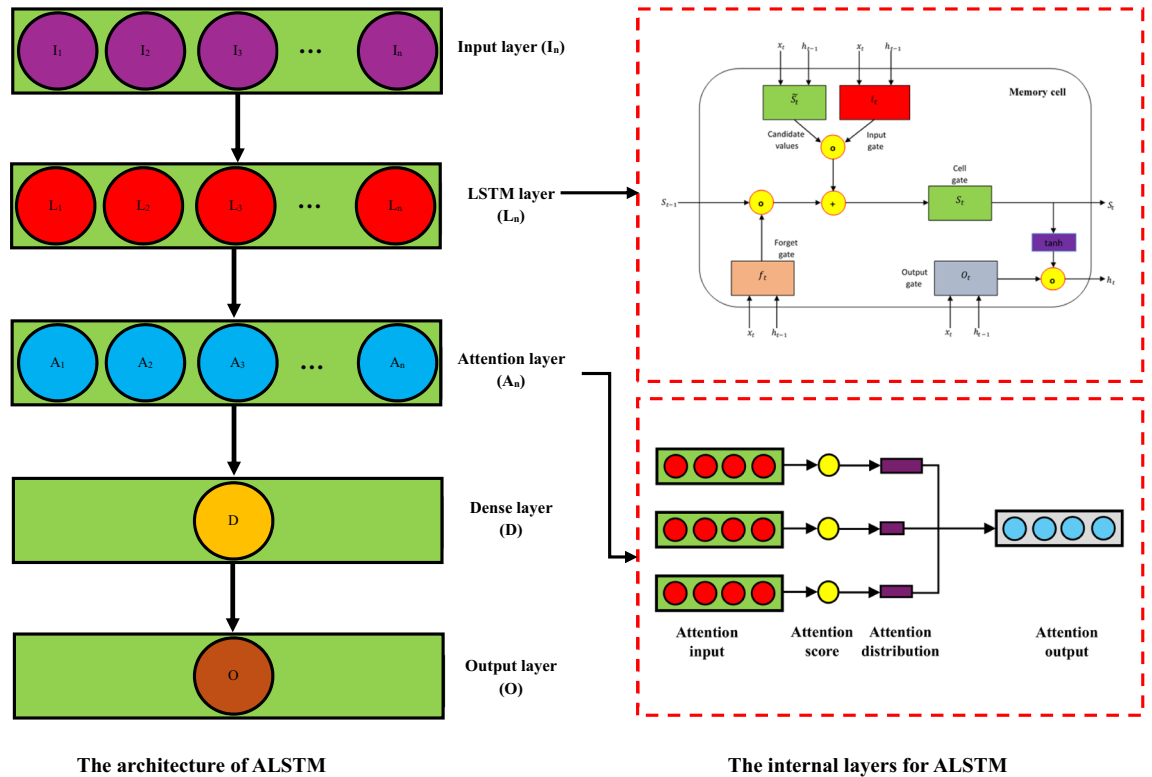


Fig. 6. The architecture of the attention layer.



The architecture of ALSTM

The internal layers for ALSTM

Fig. 7. The architecture of the ALSTM model.

$$e_{t,i} = a(S_{t-1}, h_i) \quad (20)$$

$$\alpha_{t,i} = \text{softmax}(e_{t,i}) \quad (21)$$

$$c_t = \sum_{i=1}^T \alpha_{t,i} h_i \quad (22)$$

The architecture of the ALSTM model, which integrates both an attention unit and an LSTM layer, is illustrated in Fig. 7. As illustrated, the proposed ALSTM framework is composed of five consecutive layers: an input layer, an LSTM layer, an attention layer, a dense (fully connected) layer, and an output layer. The input layer gathers raw data points and passes them to the LSTM layer, where temporal dependencies are captured through long short-term memory units that transform the input into high-level feature representations. The attention layer then generates a weight vector that highlights the most relevant data within the hidden states, allowing the model to concentrate on selectively on significant time steps while assigning lower importance to less informative ones. The dense layer connects each neuron in one layer to every neuron in the next, ensuring comprehensive feature integration before final prediction. Finally, the output layer interprets the learned representations to generate accurate time-series forecasts.

By combining the LSTM's capacity for long-term dependency modeling with the attention mechanism's ability to emphasize critical inputs, this hybrid architecture effectively mitigates the limitations of standard LSTM networks. It enhances the model's accuracy and robustness, particularly in noisy or volatile environments, by dynamically prioritizing the most informative portions of sequential data.

Hybrid model

To improve the robustness and representational capability of the prediction structure, this study employs a hybrid model architecture that integrates temporal and spatial learning mechanisms within a unified structure. The model is developed to simultaneously extract the long-term temporal relationships present in temporal financial data and the localized feature correlations among multiple input variables, thereby improving its capacity to learn complex nonlinear relationships in stock price dynamics.

In the proposed framework, sequential input vectors containing ten features per trading day including historical, technical, and sentiment-based variables are first processed by the ALSTM layer. This recurrent component maintains both hidden and cell states across time steps, enabling the network to efficiently model long-range temporal relationships within the data. To further refine the temporal representations, an attention mechanism is applied to the series of hidden states produced by the ALSTM. Through a softmax-based weighting process, the attention layer calculates a context vector that emphasizes the most informative time steps, enabling the model to concentrate on selectively on patterns that have stronger predictive significance while mitigating the vanishing gradient problem commonly encountered in long sequences.

The contextualized temporal representations obtained from the ALSTM-attention module are then passed through a two-layer one-dimensional CNN, which operates as a feature extractor. The first convolutional layer, equipped with 16 filters and a kernel size of three, identifies short-term fluctuations and localized transitions in the hidden representations. The subsequent convolutional layer, with eight filters of the same kernel size, refines these feature maps to highlight essential temporal cues. An adaptive average pooling layer subsequently condenses the learned features into a compact, high-level representation that summarizes global temporal dynamics. This CNN branch enables the network to detect abrupt variations and localized dependencies that are often overlooked by purely recurrent models.

The outputs from the attention-based ALSTM module and the pooled CNN representation are integrated to form an extensive feature vector that integrates both global sequential context and localized temporal detail. This fused representation is then fed into a completely linked dense layer responsible for predicting the next-day closing price of the stock. The model was trained with the Adam optimization algorithm at a learning rate of 0.001, and training proceeds for 50 epochs under the MSE loss criterion. The implementation is carried out in PyTorch, with computations executed on a GPU when available to ensure efficient training and evaluation. This Fig. 8 presents the structure of the introduced Hybrid model.

The rationale behind this hybrid configuration lies in the complementary strengths of its constituent components. The ALSTM component provides the capability to model long-term dependencies and capture memory-driven sequential behavior, while the CNN component improves the model's responsiveness to short-term variations and localized correlations within the input space. The attention mechanism further contributes interpretability and stability by dynamically adjusting the importance assigned to individual time steps based on their significance to the predictive task. Consequently, the hybrid model framework delivers a more robust, memory-aware, and spatially attentive architecture capable of effectively modeling the intricate and nonlinear dependencies that characterize financial time-series data.

Experiment and results

In this section, the execution process of the introduced model on the datasets corresponding to eight selected stocks is described. Initially, the hyperparameters employed in the predictive models are defined and discussed to ensure transparency and reproducibility of the trials. Subsequently, the evaluation metrics used to measure

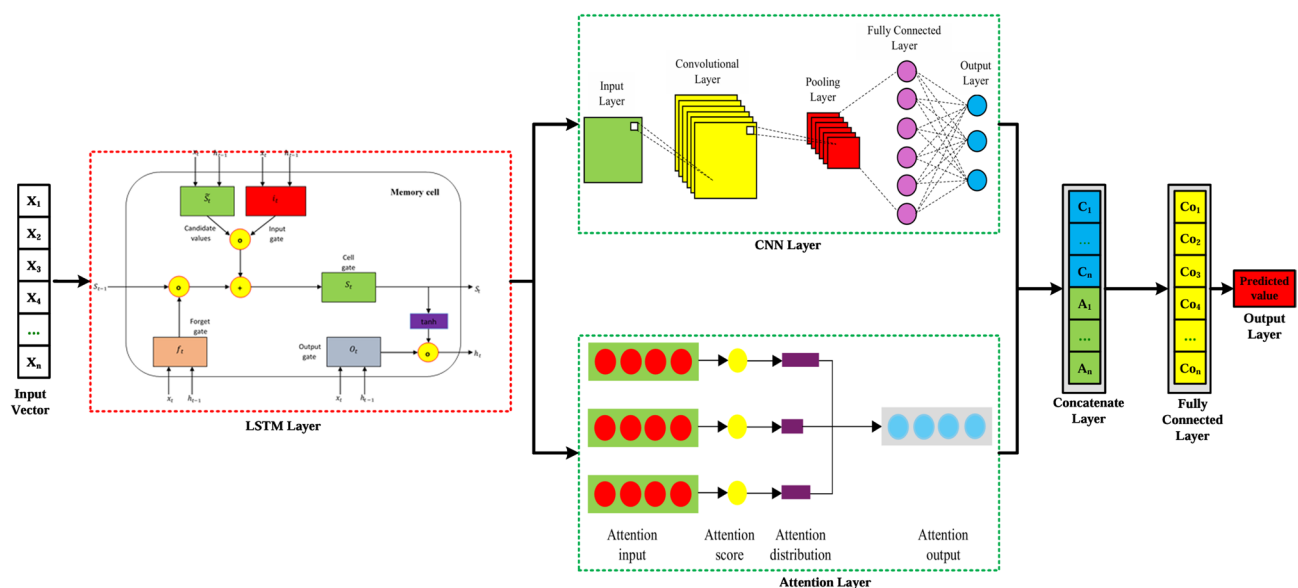


Fig. 8. The structure of the hybrid model.

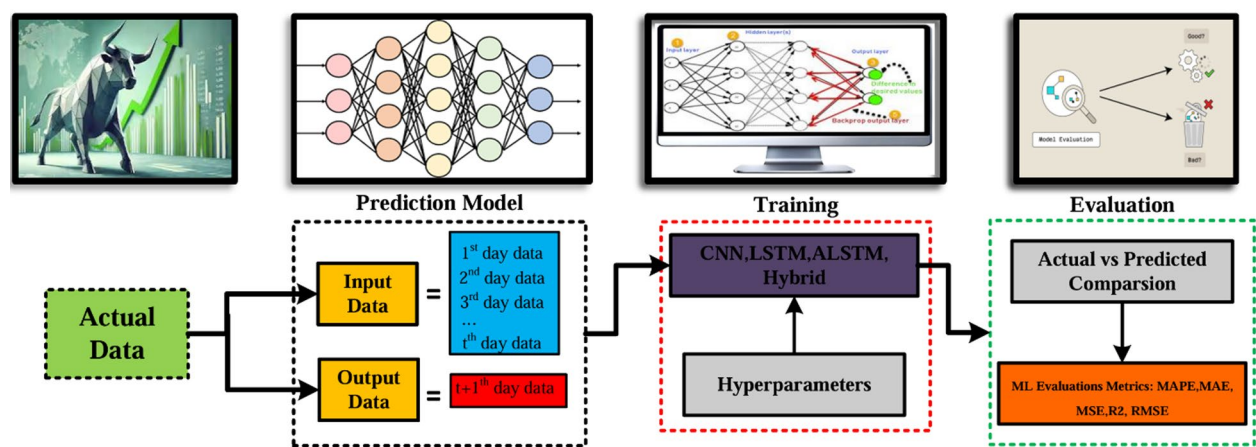


Fig. 9. The workflow of the proposed methodology.

Parameter	Value
Model type	Sequential
Input shape	(X_train.shape[1], 1)
Convolutional layer	Conv1D(filters = 64, kernel_size = 2, activation='relu')
MaxPooling layer	MaxPooling1D(pool_size = 2)
Flatten layer	Flatten()
Dense layer (Hidden)	Dense(50, activation='relu')
Dense layer (Output)	Dense(1)
Activation functions	ReLU (Conv1D, Dense), Linear (Output layer)
Optimizer	Adam
Loss function	Mean Squared Error (mean_squared_error)
Epochs	50
Batch size	32
Total trainable parameters	~ 3,300 (depends on input size)

Table 4. Summary of the selected hyperparameters and their configurations for the CNN.

model efficiency are presented and explained in detail. Finally, the experimental results obtained from applying each model to the individual stocks are reported separately, allowing for a comprehensive comparison of predictive accuracy across models and datasets. Figure 9 illustrates the overall experimental workflow used in this study for stock price prediction. The workflow starts by acquiring actual financial market data, which is divided into input and output datasets where data from previous trading day's serve as input features and the subsequent day's price represents the prediction goal. The data are then used to train four predictive models under specific hyperparameter settings. After training, the models' performance is assessed by comparing the actual and predicted stock prices based on five statistical evaluation metrics: MAPE, MAE, MSE, R², and RMSE.

In this study, the stock prices of the eight selected companies were predicted using four forecasting models over the defined one-year observation period. For each model, the dataset was split into training and testing subsets, with 80% of the data utilized for training and the remaining 20% reserved for testing to assess out-of-sample performance. Two experimental settings were designed to assess the effect of incorporating sentiment information. In the model with SA, three types of data historical stock data, technical indicators, and sentiment polarity scores were integrated, resulting in a total of 10 input features for each predictive model. In contrast, in the model without SA, only two categories of data historical data and technical indicators were used, yielding a total of 9 input features. This comparative design allowed for a systematic evaluation of the contribution of sentiment-based features to the predictive performance of the proposed models. For each of the four prediction models, specific hyperparameters were tuned to maximize the model's predictive performance. All hyperparameters were validated through five-fold cross-validation to ensure model robustness, generalization capability, and to avoid overfitting.

Hyperparameters

The detailed configuration of the selected hyperparameters for each model is summarized in the following tables: Table 4 for the CNN model, Table 5 for the LSTM model, Table 6 for the ALSTM model, and Table 7 for the Hybrid model.

Parameter	Value
Model type	Sequential
Input shape	(X_train.shape[1], 1)
LSTM layer	LSTM(units = 50, activation='relu')
Output layer	Dense(1)
Activation functions	ReLU (LSTM), Linear (Output layer)
Optimizer	Adam
Loss function	Mean Squared Error (mean_squared_error)
Epochs	50
Batch size	32
Total trainable parameters	~2,600 (depending on input size)

Table 5. Summary of the selected hyperparameters and their configurations for the LSTM.

Parameter	Value
input_size	9
hidden_size	32
seq_len	10
batch_size	32
epochs	20
optimizer	Adam
learning_rate	0.001
loss_function	MSELoss
attention_layer	Linear(hidden_dim → 1)
device	CUDA / CPU

Table 6. Summary of the selected hyperparameters and their configurations for the ALSTM.

Component	Parameter	Value
LSTM	Hidden units	32
	Batch first	TRUE
	Dropout	0
Attention	Attention dimension	32
	Attention type	Softmax + Linear
CNN	Conv1 out channels	16
	Conv2 out channels	8
	Kernel size	3
	Padding	1
	Pooling	AdaptiveAvgPool1d(1)
Fully connected	Output dimension	1
	Input size to FC layer	32 + 8 = 40
Training	Optimizer	Adam
	Learning rate	0.001
	Loss function	MSELoss
	Batch size	32
	Epochs	20
	Early stopping	Optional

Table 7. Summary of the selected hyperparameters and their configurations for the Hybrid.

CNN:

The CNN model was implemented using a sequential architecture to extract short-term relationships and local patterns in stock price movements. The input shape was defined as (X_train.shape[1], 1). The model comprises a Conv1D layer with 64 filters and a kernel size of two using ReLU activation, followed by a MaxPooling1D layer (pool size = 2) to decrease feature space and prevent overfitting. The output was flattened and passed through

a Dense hidden layer with 50 neurons (ReLU), and finally a Dense output layer with linear activation for price forecasting. The model was trained with the Adam optimization algorithm and MSE as the loss function for 50 epochs, using a batch size of 32. Hyperparameters such as filter number, kernel size, and hidden units were tuned using five-fold cross-validation to achieve optimal accuracy. The total trainable parameters were approximately 3,300, depending on the input dimension.

LSTM:

The LSTM model was implemented using a sequential architecture to efficiently extract long-term temporal relationships in stock price data. The input structure was defined as $(X_train.shape[1], 1)$, representing the sequence length and single-feature dimension per timestep. The model consists of a single LSTM layer with 50 memory units and a ReLU activation function to introduce nonlinearity and accelerate convergence. The output layer is a fully connected dense neuron with linear activation, suitable for continuous value prediction. The model was trained with the Adam optimization algorithm and the MSE loss function for 50 epochs, using a batch size of 32. Hyperparameters such as the number of LSTM units, learning rate, and batch size were selected through empirical experimentation and five-fold cross-validation, to achieve a trade-off between model accuracy and overfitting mitigation. The final configuration resulted in approximately 2,600 trainable parameters, depending on the input dimension, providing a compact yet powerful model for time-series forecasting.

ALSTM:

The ALSTM model was designed to improve the interpretability and precision of sequential learning by assigning adaptive importance weights to each timestep in the input series. The model receives input vectors of size 9 features across 10-time steps ($seq_len = 10$) and processes them through an LSTM layer with a hidden size of 32, enabling it to capture long-term temporal relationships. An attention layer, implemented as a linear transformation mapping from the hidden dimension to a single scalar weight, computes the relative contribution of each hidden state to the final context vector. This mechanism enables the model to concentrate on selectively on the most informative patterns within the sequence.

The network was trained over 20 epochs with a mini-batch size of 32, employing the Adam optimizer and MSE loss function. The learning rate was set to 0.001, ensuring stable and efficient convergence. Training and evaluation were performed on both CUDA-enabled GPU and CPU, depending on availability. All hyperparameters were fine-tuned through five-fold cross-validation, ensuring an optimal trade-off between learning performance and generalization capability.

Hybrid: The proposed Hybrid model was designed to integrate the strengths of recurrent and convolutional architectures for enhanced stock price prediction. The LSTM component was configured with 32 hidden units, operating in batch-first mode without dropout to preserve sequential integrity. An attention mechanism was incorporated with a 32-dimensional attention space, using a softmax-linear combination to assign adaptive weights to temporal features, thus allowing the model to attend to the most informative time steps within each sequence.

The CNN branch consisted of two one-dimensional convolutional layers: the first with 16 output channels and the second with 8, both using a kernel size of 3 and padding of 1 to preserve temporal resolution. An adaptive average pooling layer ($AdaptiveAvgPool1d(1)$) was applied to condense feature maps into a compact representation, capturing localized temporal dependencies. The concatenated output from the ALSTM-attention and CNN modules (with a combined input size of 40 features) was passed into a fully connected layer producing a single scalar output corresponding to the predicted stock price.

The model was trained with the Adam optimizer (learning rate = 0.001) using the MSE loss function over 20 epochs with a batch size of 32. An early stopping strategy was optionally employed to prevent overfitting. Hyperparameters were fine-tuned through five-fold cross-validation, ensuring a balance between learning efficiency and generalization accuracy across all stock datasets.

Hyperparameter optimization

The hyperparameters for all models were optimized through empirical experimentation and five-fold cross-validation. Parameters such as the number of filters, hidden units, learning rate, and batch size were systematically tuned to minimize validation loss while preventing overfitting. The Adam optimizer was employed across all architectures due to its adaptive learning capability, and the MSE loss function was used. Epochs and batch sizes were adjusted to balance computational efficiency and model stability, as the objective criterion. The number of epochs and batch size were tuned to achieve a trade-off between computational efficiency and model stability. This systematic approach ensured that each model operated at its optimal configuration, providing a fair and consistent basis for performance comparison across all prediction frameworks.

Evaluation metrics

The effectiveness evaluation of the four prediction models in this study considering both with and without SA scenarios for the eight selected stocks relies on five statistical evaluation metrics. These measures collectively offer an extensive understanding of the models' predictive accuracy and robustness from different analytical perspectives. The subsequent subsections provide a detailed explanation of each evaluation metric.

Mean squared error

The MSE is an extensively utilized assessment metric that quantifies a model's forecasting precision by calculating the mean of the squared discrepancies between estimated and observed values. It acts as a straightforward and understandable measure of overall model effectiveness, with greater values indicating increased prediction errors. Squaring the residuals in MSE increases the effect of larger deviations, which makes it particularly appropriate for scenarios where significant errors are more crucial and need to be penalized more severely. Equation (23) presents the mathematical formulation for calculating the MSE:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (23)$$

Where:

- N represents the total amount of observations.
- y_i is the observed value for the i th observation.
- $\hat{y}_i$ is the estimated value for the i th observation.

Mean absolute error

The MAE is a commonly employed evaluation metric that quantifies the average absolute deviation between predicted and actual values, providing a simple assessment of a model's forecasting precision. In contrast to MSE which disproportionately penalizes larger deviations by squaring them MAE treats all errors equally by taking their absolute values before averaging. Owing to its simplicity, interpretability, and resilience to outliers, MAE serves as a reliable and informative indicator in both regression and forecasting analyses. Equation (24) provides the mathematical expression for computing the MAE:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (24)$$

Where:

- N represents the total amount of observations.
- y_i is the observed value for the i -th observation.
- $\hat{y}_i$ is the estimated value for the i -th observation.

Root mean squared error

The RMSE is a well-established performance indicator commonly applied in regression assessment. It measures the mean magnitude of the differences between estimated and observed values, giving greater weight to larger deviations between predicted and actual values. RMSE is computed as the square root of the mean of the squared residuals, offering a direct interpretation of how far predictions typically fall from true observations. Because of its sensitivity to large errors, RMSE serves as a valuable metric in scenarios where substantial deviations can significantly affect model reliability and decision-making accuracy. Equation (25) presents the mathematical formulation for calculating the RMSE:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (25)$$

Where:

- N represents the total amount of observations.
- y_i is the observed value for the i th observation.
- $\hat{y}_i$ is the estimated value for the i th observation.

Mean absolute percentage error

The MAPE is an extensively adopted indicator for assessing the predictive performance of prediction models. It measures the average percentage deviation between predicted and actual values over all observations, providing an interpretable and scale-independent assessment of forecasting accuracy. Equation (26) provides the mathematical representation used to compute the MAPE:

$$MAPE = 100 \times \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (26)$$

Where:

- N represents the total amount of observations.
- y_i is the observed value for the i -th observation.
- $\hat{y}_i$ is the estimated value.

R^2

The R^2 is a widely utilized statistical measure that indicates how effectively a regression model captures the fluctuation in the observed data. It expresses the ratio of total variance in the reliant on variable that is explained by the model's independent variables, thereby reflecting the model's descriptive strength. Although R^2 offers meaningful insights into the overall fit of a model, it must be interpreted carefully especially when comparing models that incorporate varying numbers of predictors or nonlinear structures since its value can be affected by both model complexity and the underlying characteristics of the dataset. Equation (27) presents the mathematical formulation for calculating the R^2 :

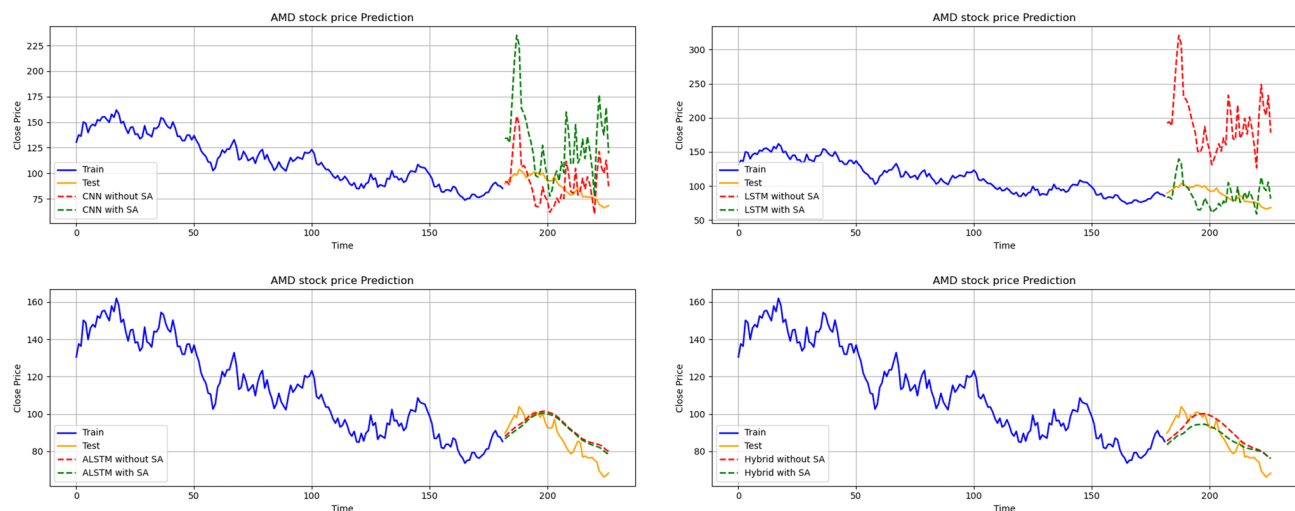


Fig. 10. Predicted and actual closing prices for AMD stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	67.909	7.046	8.241	0.445	0.087	60.978	6.754	7.809	0.502	0.083
CNN	580.919	19.344	24.102	-3.744	0.227	3100.518	44.182	55.682	-24.320	0.526
Hybrid	47.038	5.877	6.858	0.616	0.071	46.061	5.798	6.787	0.624	0.068
LSTM	12911.998	105.326	113.631	-104.442	1.238	442.141	17.528	21.027	-2.611	0.204

Table 8. Comparison of the predictive performance of four models using five evaluation metrics for AMD.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (27)$$

Where:

- N represents the total amount of observations.
- y_i is the observed value for the i -th observation.
- $\hat{y}_i$ is the estimated value for the i -th observation.
- $\bar{y}_i$ is the mean of the actual values of observations.

Results

This subsection provides the results derived from implementing the predictive models on each of the selected stocks. The effectiveness of the models is evaluated using the previously defined evaluation metrics, allowing for a comparative analysis between the models with and without sentiment integration.

Result of AMD

Figure 10 illustrates the actual and predicted closing prices of the AMD stock over the observation period, comparing the results of the prediction model with SA and without SA.

As shown in the Fig. 10, the model incorporating SA exhibits predictions that are visually closer to the actual stock prices, demonstrating its enhanced ability to capture market fluctuations and price dynamics. This visual alignment highlights the model's enhanced ability to extract short-term volatility, market fluctuations, and underlying price dynamics influenced by investor sentiment and behavioral factors. In addition to the visual comparison, the quantitative results summarized in Table 8 confirm this improvement. Across all four predictive models, the inclusion of sentiment polarity scores led to lower error metrics and higher R^2 values, indicating a significant enhancement in forecasting accuracy and robustness. Overall, both the graphical and numerical analyses confirm that integrating sentiment data as an additional feature increases the performance of stock price forecasting models by allowing them to better interpret the behavioral and psychological aspects of market movements.

As presented in Table 8, the introduced Hybrid model shown the best overall forecasting accuracy among all four models evaluated. The integration of both CNN and recurrent ALSTM layers allowed the model to efficiently capture both spatial relationships within the input features and temporal patterns across time

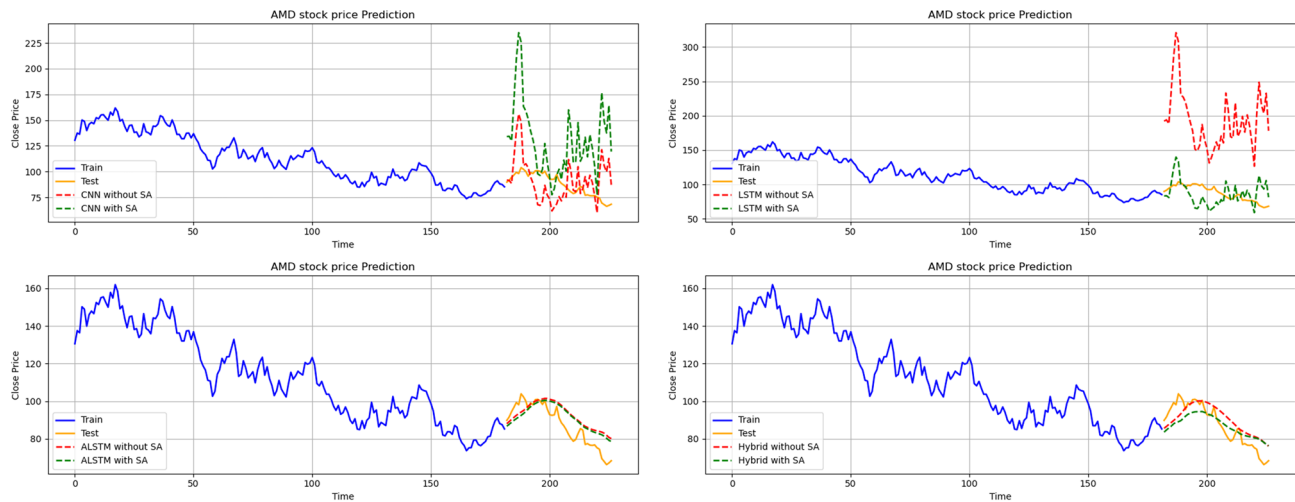


Fig. 11. Predicted and actual closing prices for AMZN stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	105.903	7.873	10.291	-0.334	0.058	79.131	7.737	8.896	0.003	0.060
CNN	2564.650	43.649	50.642	-31.313	0.330	19051.542	137.700	138.027	-239.041	1.056
Hybrid	71.383	6.578	8.449	0.101	0.049	84.582	7.084	9.197	-0.066	0.053
LSTM	3040.286	51.013	55.139	-37.306	0.384	2548.798	45.111	50.486	-31.114	0.339

Table 9. Comparison of the forecasting performance of four models using five evaluation metrics for AMZN.

sequences. This combination enhanced the model’s capability to extract richer representations from the data, leading to more accurate forecasts of stock prices. Quantitatively, the Hybrid model achieved lower MSE, MAE, RMSE, and MAPE values and a higher R^2 compared to the CNN, LSTM, and ALSTM models. These results confirm that the proposed architecture not only benefits from the inclusion of sentiment features but also utilizes the complementary strengths of CNN and LSTM networks. Consequently, the Hybrid model provides a more robust and precise framework for stock price prediction, demonstrating its superiority in capturing both technical and behavioral market dynamics.

Result of AMZN

Figure 11 illustrates the actual and predicted closing prices for AMZN stock using the Prediction models with and without SA.

The Fig. 11 demonstrates that in the case of AMZN, the results depicted in the figure clearly illustrate the efficacy of integrating SA into the prediction model. The model enhanced with sentiment information demonstrates a much stronger alignment with the actual stock price movements compared to the version without sentiment integration. During the testing phase, it successfully captures the direction and magnitude of short-term fluctuations, particularly around periods of sudden price reversals and local peaks. This improvement can be attributed to the influence of investor sentiment derived from Twitter data, which provides valuable behavioral insights into market dynamics. The quantitative comparison in Table 9 further supports this observation. For the ALSTM model, the inclusion of sentiment polarity scores reduced the MSE from 105.90 to 79.13, the RMSE from 10.29 to 8.89, and slightly improved the R^2 value from -0.33 to 0.00 , reflecting better predictive accuracy and reduced deviation from actual values. Although the improvement magnitude varies among models, the Prediction models with SA clearly outperforms its counterpart without SA in terms of precision and consistency. Visually and statistically, these results confirm that the incorporation of sentiment-based features improves the model’s capability to interpret market reactions and public opinion, resulting in to more realistic and precise stock price forecasts.

As observed in Table 9, the proposed Hybrid model attained the best overall efficiency among all four predictive models for AMZN stock. The combination of CNN for feature extraction and recurrent layers (ALSTM) for temporal learning allowed the Hybrid model to effectively extract both short-term variations and long-term dependencies in price dynamics. Quantitatively, the Hybrid model achieved the lowest MAE among all models in the scenario without SA, with a value of 6.58, compared to 7.87 for ALSTM, 43.65 for CNN, and 51.01 for LSTM. This result indicates that, on average, the Hybrid model’s predicted prices deviated less from the actual prices, reflecting its superior precision. Moreover, the Hybrid model also exhibited the highest R^2 value (0.10) in the same configuration, confirming its stronger explanatory power in modeling stock price variations.

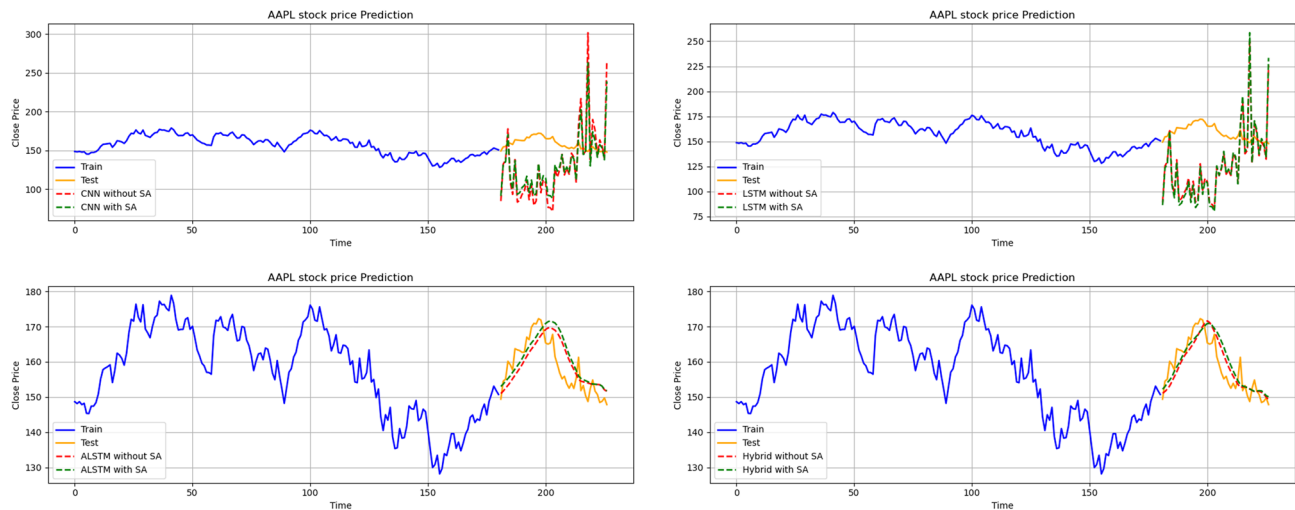


Fig. 12. Predicted and actual closing prices for AAPL stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	27.583	4.640	5.252	0.479	0.029	30.496	4.583	5.522	0.424	0.029
CNN	3395.635	47.699	58.272	-63.190	0.298	2416.445	41.005	49.157	-44.679	0.255
Hybrid	15.720	3.333	3.965	0.703	0.021	17.359	3.357	4.166	0.672	0.021
LSTM	2559.910	43.183	50.596	-47.391	0.269	2838.542	44.965	53.278	-52.659	0.280

Table 10. Comparison of the forecasting performance of four models using five evaluation metrics for AAPL.

These findings demonstrate that the Hybrid model not only benefits from the integration of multiple data sources (historical, technical, and sentiment-based features) but also from its combined network architecture, which effectively captures complex nonlinear relationships in financial time series data.

AAPL

Figure 12 illustrates the actual and predicted closing prices for AAPL stock using the Prediction models with and without SA. As can be observed, the predictions obtained from the model incorporating SA are visually more aligned with the actual closing prices, particularly in the testing segment.

Figure 12 suggests that integrating public sentiment information allows the model to better capture the behavioral dynamics influencing Apple’s stock movements. The quantitative findings presented in Table 10 further confirm this observation. For the ALSTM model, the MAE slightly decreased from 4.64 to 4.58, and the MAPE improved from 0.0292 to 0.0290, while maintaining a relatively high R^2 value of 0.42, compared to 0.47 in the version without sentiment. Although the improvement margin for AAPL is smaller contrasted to other stocks, these results still show that the integration of sentiment data contributes to more consistent and realistic forecasts. Overall, both the graphical representation and the numerical evaluation demonstrate that models incorporating SA tend to produce smoother and closer-fitting predictions, capturing subtle market reactions to investor sentiment that are not reflected in historical and technical data alone. Table 10 indicates that the introduced Hybrid model achieved the best overall efficiency relative to the other three predictive models in predicting AAPL stock.

Based on the reported results in Table 10, the Hybrid model once again demonstrated a superior predictive capability compared to the other three models. The advantage of this architecture lies in its capability to integrate feature extraction through convolutional layers with temporal sequence learning via the ALSTM component, which together enable a more extensive understanding of complex stock price behaviors. For the AAPL stock, the Hybrid model achieved the lowest MAE of 3.33 and the highest R^2 value of 0.70 in the scenario without SA, outperforming both the ALSTM model (MAE = 4.64, R^2 = 0.48) and the LSTM model (MAE = 43.18, R^2 = -47.39). These values show that the Hybrid model provided more precise and stable forecasts, with significantly reduced deviation from the actual prices. Even after incorporating SA, the Hybrid model maintained its superiority, achieving consistent error reduction and stable explanatory power. This evidence highlights that the Hybrid approach not only benefits from sentiment information but also effectively captures multi-dimensional dependencies within the financial time series, leading to more reliable and nuanced forecasting outcomes.

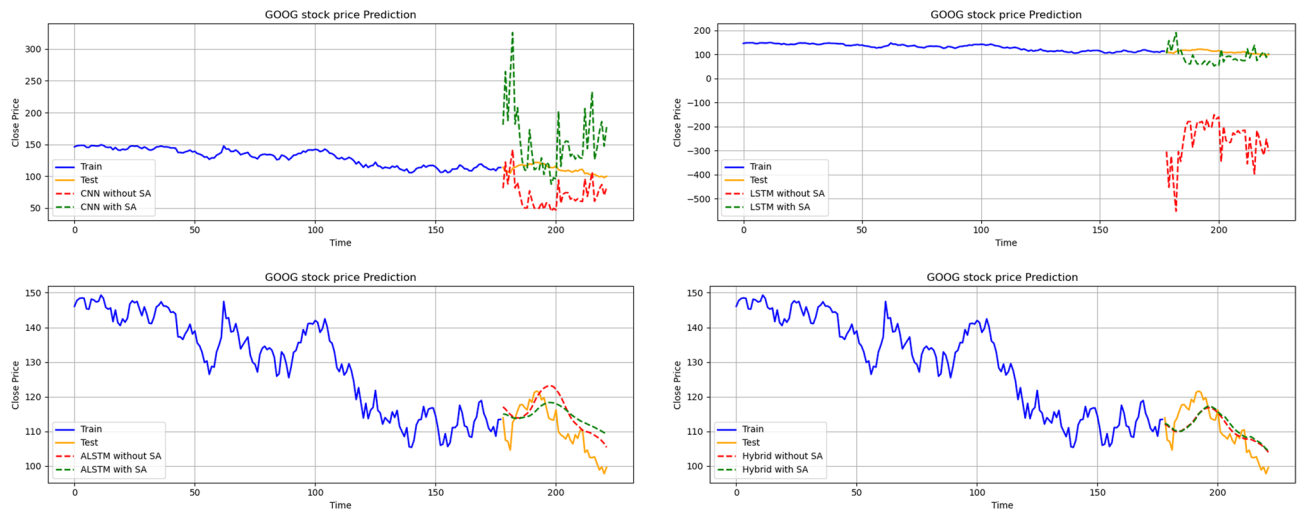


Fig. 13. Estimated and observed closing prices for GOOG stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	40.509	5.453	6.365	0.099	0.050	42.784	5.735	6.541	0.049	0.053
CNN	2145.449	41.814	46.319	-46.699	0.371	4582.711	49.146	67.696	-100.886	0.456
Hybrid	20.833	4.121	4.564	0.537	0.037	24.045	4.487	4.904	0.465	0.041
LSTM	145087.632	371.955	380.904	-3224.678	3.389	1523.385	33.258	39.031	-32.869	0.295

Table 11. Comparison of the predictive performance of four models using five evaluation metrics for GOOG.

GOOG

Figure 13 illustrates the comparing the prediction models with and without SA for the actual and predicted closing prices for GOOG stock.

As illustrated, the prediction lines of the model with SA follows the actual trend more closely, especially in the latter phase of the testing period, capturing both short-term declines and minor recoveries with improved precision. This behavior indicates that sentiment information derived from Twitter data adds a behavioral dimension to the model, allowing it to better interpret market responses to investor emotions.

The performance metrics reported in Table 11 further confirm this observation. For the ALSTM model, the incorporation of SA slightly reduced the R^2 value from 0.099 to 0.049 but produced a more stable forecast pattern in the testing phase. More notably, across all models, the Hybrid network once again outperformed the others. It achieved the lowest MSE (20.83) and lowest MAE (4.12) in the scenario without SA, indicating superior accuracy and stability in predicting GOOG's price movements. In comparison, the ALSTM model's MSE and MAE were significantly higher (40.50 and 5.45, respectively), highlighting the robustness of the Hybrid structure. Overall, both the visual and quantitative analyses demonstrate that integrating SA improves the ability of the models to track real market dynamics, while the Hybrid model remains the most effective and resilient approach for capturing the nonlinear and multidimensional nature of financial time series data.

In light of the obtained results in Table 11, the Hybrid model clearly showed superior predictive efficiency contrasted to the other models for GOOG stock. The improved performance primarily stems from the model's integrated architecture, which integrates the feature extraction ability of CNN layers with the temporal learning capability of the ALSTM module. Such a configuration enables the model to simultaneously capture local feature patterns in the data and long-term dependencies across time, resulting in more accurate and stable predictions. Numerically, the Hybrid model achieved an MSE of 20.83 and an MAE of 4.12 in the setting without SA, both of which are significantly lower than those of the ALSTM model (MSE = 40.51, MAE = 5.45). Furthermore, the R^2 value of 0.53 obtained by the Hybrid model indicates a stronger ability to explain price variations compared to all other models. Even after incorporating SA, the Hybrid model maintained a consistent level of accuracy (MSE = 24.04, MAE = 4.49), showing that its predictive strength is robust against different feature configurations. These results reaffirm that the Hybrid architecture, by leveraging both convolutional and recurrent mechanisms, offers a more thorough representation of financial time series data, allowing it to outperform single-structure models in representing the intricate and nonlinear dynamics of stock market behavior.

MSFT

The predictive results for MSFT stock are introduced in Fig. 14; Table 12, illustrating the model performance with and without SA. As can be observed, incorporating sentiment data slightly improved the model's capability

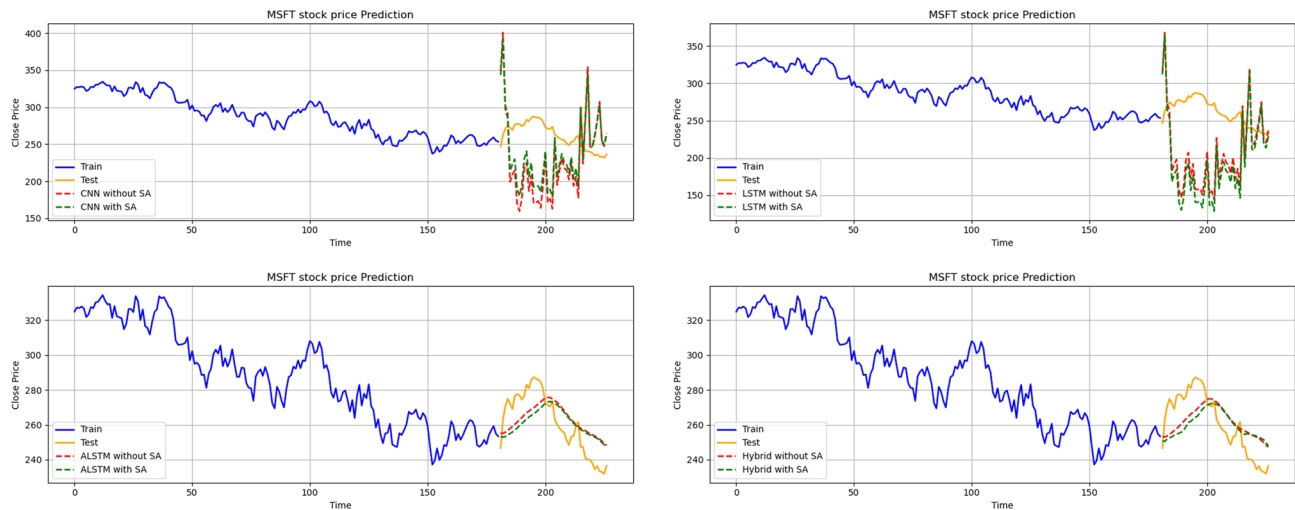


Fig. 14. Predicted and actual closing prices for MSFT stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	173.838	12.289	13.185	0.417	0.048	200.686	12.943	14.166	0.327	0.050
CNN	5413.460	63.745	73.576	-17.164	0.240	3905.298	54.324	62.492	-12.103	0.205
Hybrid	182.528	12.469	13.510	0.388	0.048	229.052	13.631	15.134	0.231	0.052
LSTM	6665.539	70.399	81.643	-21.365	0.262	8552.165	80.240	92.478	-27.695	0.299

Table 12. Comparison of the predictive performance of four models using five evaluation metrics for MSFT.

to follow the observed price trend, especially in the declining phase of the test period. The prediction models with SA produced a smoother prediction curve that remained closer to the real stock values, reflecting a better understanding of investor sentiment and market reactions.

However, when examining the numerical outcomes in Table 12, it becomes evident that the Hybrid model maintained the most balanced and reliable efficiency overall. While the inclusion of SA did not drastically reduce errors across all models, the Hybrid network continued to outperform the others in consistency and accuracy. For instance, in the configuration without SA, the Hybrid model attained a MSE of 182.53 and an R^2 value of 0.39, which are both superior to those of CNN (MSE = 5413.46, R^2 = -17.16) and LSTM (MSE = 6665.53, R^2 = -21.36).

These findings suggest that, although SA offers marginal improvements for MSFT likely due to the stock's relatively stable behavior and reduced volatility compared to others the Hybrid model remains the most robust framework. Its combined convolutional and recurrent architecture enables it to efficiently extract both short-term price movements and long-term temporal relationships, ensuring reliable predictions even under less sentiment-sensitive market conditions.

PG

The predictive results for PG stock are depicted in Fig. 15 and detailed in Table 13. As shown in the figure, the incorporation of SA led to slightly smoother and more responsive predictions in the testing phase, where the prediction models with SA was able to follow short-term upward and downward movements more accurately than the version without SA. This improvement reflects the positive influence of integrating public sentiment information derived from investor discussions on social media into the predictive process.

However, a closer look at the numerical results in Table 13 indicates that the performance gains varied across models. The Hybrid model once again demonstrated the strongest overall accuracy and stability. In the configuration without SA, it achieved the lowest MSE (14.09) and lowest MAE (2.99), with an R^2 value of 0.34, all of which surpass the results of the ALSTM model (MSE = 20.53, MAE = 3.65, R^2 = 0.04). Even after the inclusion of sentiment features, the Hybrid model retained its relative advantage, confirming its robustness across different data combinations.

These results suggest that while SA enhances short-term predictive responsiveness, the Hybrid model's integrated CNN–ALSTM architecture provides the most consistent and reliable framework for capturing both trend and volatility in the PG stock. Its ability to process spatial and temporal dependencies simultaneously enables more accurate modeling of price behavior, even in less volatile consumer goods markets such as PG.

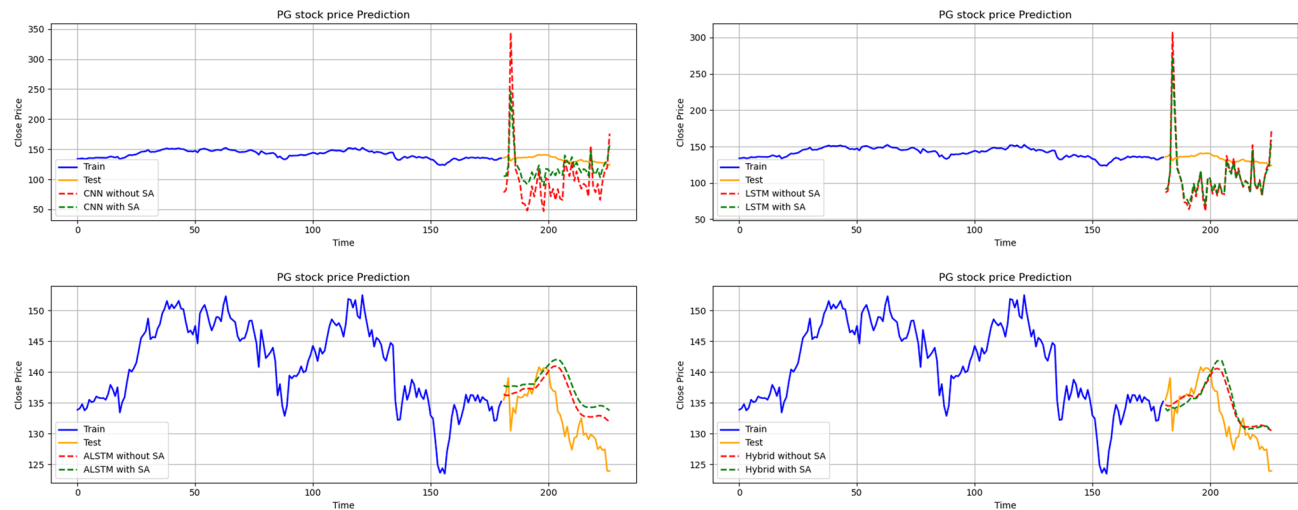


Fig. 15. Predicted and actual closing prices for PG stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	20.531	3.656	4.531	0.044	0.028	32.377	4.643	5.690	-0.508	0.036
CNN	3592.682	49.580	59.939	-166.287	0.370	921.608	24.025	30.358	-41.913	0.179
Hybrid	14.087	2.997	3.753	0.344	0.023	18.221	3.356	4.269	0.152	0.025
LSTM	2370.459	39.872	48.687	-109.376	0.297	1871.017	36.597	43.255	-86.121	0.273

Table 13. Comparison of the predictive performance of four models using five evaluation metrics for PG.

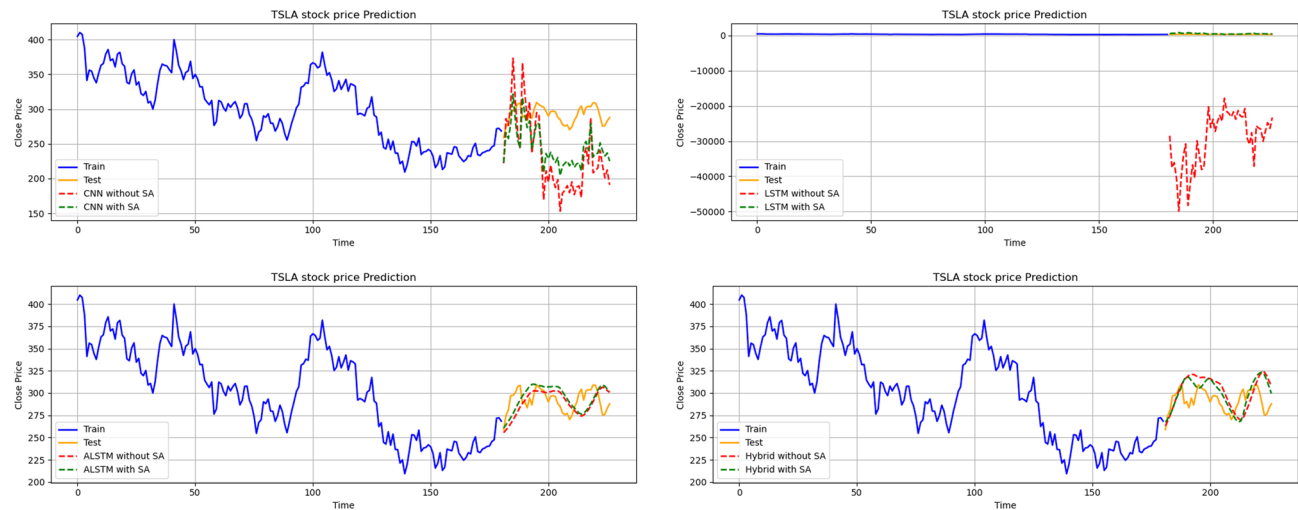


Fig. 16. Predicted and actual closing prices for Tesla stock.

TSLA

The prediction outcomes for TSLA stock, illustrated in Fig. 16 and detailed in Table 14, reveal a distinctive pattern compared to the other analyzed stocks. Due to Tesla’s highly volatile and sentiment-driven market behavior, the integration of SA demonstrates a stronger effect on the model’s forecasting ability. As shown in the figure, the ALSTM model with SA more accurately follows the fluctuations of the actual closing prices, particularly during sharp upward and downward shifts in the testing period. This suggests that Tesla’s price movements are more sensitive to investor emotions and public discourse, making sentiment data a valuable predictive feature.

From a quantitative perspective, the results in Table 14 confirm this trend. The Hybrid model achieved the most reliable balance between error reduction and trend detection. After incorporating SA, the Hybrid model’s

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	291.309	14.307	17.068	-0.995	0.049	290.184	15.077	17.035	0.001	0.052
CNN	5984.065	68.191	77.357	-39.978	0.234	10493.703	94.316	102.439	-48.859	0.323
Hybrid	445.527	17.834	21.108	-1.051	0.062	339.446	14.765	18.424	0.167	0.051
LSTM	6289.23	56.74	79.29	-1.431	12.81	4783.05	49.52	69.13	-1.512	11.02

Table 14. Comparison of the predictive performance of four models using five evaluation metrics for TSLA.

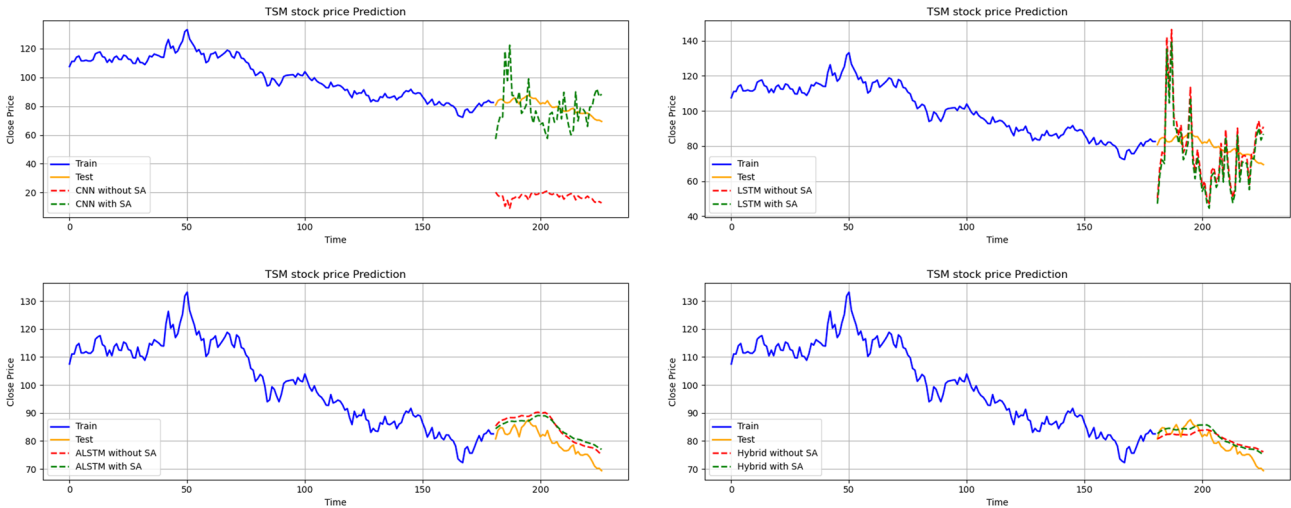


Fig. 17. Predicted and actual closing prices for TSM stock.

Model	Without sentimental					With sentimental				
	MSE	MAE	RMSE	R2	MAPE	MSE	MAE	RMSE	R2	MAPE
ALSTM	26.363	4.761	5.134	-0.104	0.060	25.849	4.583	5.084	-0.083	0.059
CNN	3971.420	62.848	63.019	-165.369	0.787	209.861	11.950	14.487	-7.791	0.150
Hybrid	11.058	2.848	3.325	0.537	0.037	8.366	2.440	2.892	0.650	0.032
LSTM	446.522	16.591	21.131	-1.706	0.208	445.384	17.214	21.104	-1.658	0.215

Table 15. Comparison of the predictive performance of four models using five evaluation metrics for TSM.

MSE decreased from 445.52 to 339.45, and its RMSE improved from 21.10 to 18.42, indicating a noticeable enhancement in predictive accuracy. Similarly, the ALSTM model also exhibited a slight improvement, reducing MSE from 291.31 to 290.18.

These improvements, though modest numerically, reflect meaningful stability in prediction given Tesla’s inherent price volatility. Overall, these findings highlight that sentiment-aware models particularly the Hybrid CNN–ALSTM architecture are more effective in forecasting emotionally reactive stocks like TSLA. The ability to fuse technical, historical, and textual sentiment information enables the model to better interpret sudden market swings driven by public attention and investor psychology.

TSM

The predictive performance for TSM stock is presented in Fig. 17; Table 15, highlighting the effect of incorporating SA into the forecasting models. As depicted in the figure, the prediction models with SA produces a prediction line that more closely follows the actual price trajectory, particularly toward the end of the testing period. This improvement indicates that including investor sentiment data helped the model adjust more effectively to downward market movements and short-term fluctuations.

A detailed examination of the results in Table 15 reveals that the Hybrid model once again outperformed all other approaches, demonstrating both precision and stability. Specifically, when SA was included, the Hybrid model attained an MSE of 8.36 and an MAE of 2.44, showing a clear improvement compared to its version without SA (MSE=11.05, MAE=2.85). Furthermore, its R² value rose from 0.54 to 0.65, signifying stronger explanatory power and a better fit to the actual price data.

These results highlight two key insights: first, sentiment information enhances the model's sensitivity to real-time market reactions, and second, the Hybrid CNN–ALSTM architecture effectively combines pattern recognition and temporal dependency modeling. This synergy allows it to deliver highly accurate and adaptive forecasts, particularly in stocks like TSM where both technical trends and investor sentiment significantly influence price behavior.

Model interpretability analysis

Given that explainable artificial intelligence has become increasingly important in financial forecasting due to model transparency for practical adoption, risk management, and regulatory compliance, the SHAP framework was employed to quantify the contribution of each input feature to the model's predictions. SHAP provides a unified, theory-based metric for feature importance by decomposing each prediction into additive feature-level contributions⁸¹. The results obtained from implementing this method is shown in Fig. 18.

As shown in Fig. 18, the SHAP analysis indicates that feature importance is not uniform across different stocks, and the contribution pattern of variables varies depending on the asset under consideration. Nevertheless, several technical indicators, particularly Williams %R, MACD, and RSI, exhibit relatively higher importance across most stocks and play a more prominent role in the model's predictions compared with other features. For instance, Williams %R shows the highest SHAP values for several stocks, including PG, TSLA, and TSM, highlighting its strong influence on the model's decision-making for these assets. Similarly, MACD has a notable contribution in stocks such as AMD, APPL, and PG, while RSI appears particularly influential in cases such as AMD. In contrast, features related to trading volume, EMA, MA, and especially the sentiment_score show comparatively lower contributions in this analysis. However, the relatively smaller importance of the sentiment feature does not imply that it is ineffective; in financial forecasting, even modest improvements can be practically meaningful and valuable. As demonstrated in the previous sections, incorporating sentiment features alongside price-based and technical variables led to improved predictive performance. Therefore, although the sentiment_score contributes less directly than some dominant technical indicators in explaining the model's predictions, it still serves as a complementary source of information that can enhance the overall forecasting performance.

Statistical significance testing

To further assess whether the observed performance differences among forecasting models are statistically meaningful, we employed the Diebold–Mariano (DM) test⁸². This test is widely used to compare the predictive accuracy of competing forecasting models based on their forecast errors. To rigorously assess whether the superior predictive performance of the proposed Hybrid model is statistically significant, the DM test was conducted across all stocks and competing models. The detailed pairwise comparison results are reported in



Fig. 18. Heatmap of SHAP feature importance across companies.

Table 16, where the forecasting accuracy of the Hybrid model is evaluated against ALSTM, CNN, and LSTM for each stock.

As shown in Table 16, the Hybrid model achieves statistically significant improvements ($p < 0.05$) in 21 out of 24 comparisons, corresponding to 87.5% of all cases, as summarized in the Summary results. This high proportion of significant outcomes provides strong evidence that the observed performance gains of the Hybrid model are not due to random fluctuations but reflect a consistent and reliable improvement in predictive accuracy. A stock-level analysis further confirms the robustness of the Hybrid model. For several stocks, including APPL, GOOG, MSFT, PG, and TSM, the Hybrid model significantly outperforms all baseline models in every comparison. In contrast, only a limited number of cases—specifically Hybrid versus ALSTM in AMD, AMZN, and TSLA—show non-significant differences, indicating comparable performance in these instances rather than inferiority of the Hybrid approach.

In addition, Fig. 19 presents an aggregated heatmap of the average p-values obtained from the DM tests across all stocks. The heatmap reveals that comparisons involving the Hybrid model, particularly against CNN and LSTM, consistently yield extremely low p-values (on the order of 10^{-6} to 10^{-5}), highlighting strong statistical significance. Although the average p-value for Hybrid versus ALSTM is relatively higher, the overall results still favor the Hybrid model in the majority of individual comparisons.

The cumulative heatmap in Fig. 19 shows the average p-values of pairwise comparisons, where statistically significant differences ($p < 0.05$) are highlighted. The results indicate that the hybrid model significantly outperforms the baseline models in most cases, demonstrating consistent superiority across various market conditions. Overall, the combined evidence from Table 16; Fig. 19 clearly demonstrates that the proposed Hybrid model delivers statistically significant improvements over traditional deep learning models. The consistency of these findings across multiple stocks underscores the robustness and generalizability of the Hybrid architecture for financial time-series forecasting.

Discussion

In this study, we proposed a hybrid model for stock price prediction, which collected and integrated three distinct types of historical stock data, technical indicator data, and sentiment text data to improve the prediction accuracy of stock price prediction models. We used the FinGPT model to analyze sentiment on the collected texts. The dataset covered a one-year period for eight major stocks. Historical data, including open, close, high, low, and volume, were obtained from Yahoo Finance, while technical indicators were calculated based on closing prices. Sentiment text data were collected from Twitter using Kaggle's Sunscap tool and processed to extract daily sentiment polarity scores for each stock. Four DL models were used for price prediction: CNN, LSTM, ALSTM, and a hybrid model. Each model was trained and evaluated using an 80/20 split for training and testing the dataset. Two experimental scenarios were considered: one with SA and one without SA. In the

Stock	Model-1	Model-2	DM-Stat	p-value	Significant
AMD	Hybrid	ALSTM	-1.9142	0.055595	0
AMD	Hybrid	CNN	-4.96187	6.98E-07	1
AMD	Hybrid	LSTM	-5.62156	1.89E-08	1
AMZN	Hybrid	ALSTM	0.266965	0.789496	0
AMZN	Hybrid	CNN	-46.2328	0	1
AMZN	Hybrid	LSTM	-7.48351	7.24E-14	1
APPL	Hybrid	ALSTM	-4.28295	1.84E-05	1
APPL	Hybrid	CNN	-6.07081	1.27E-09	1
APPL	Hybrid	LSTM	-6.79293	1.1E-11	1
GOOG	Hybrid	ALSTM	-3.73606	0.000187	1
GOOG	Hybrid	CNN	-3.57837	0.000346	1
GOOG	Hybrid	LSTM	-6.67752	2.43E-11	1
MSFT	Hybrid	ALSTM	4.020355	5.81E-05	1
MSFT	Hybrid	CNN	-6.54269	6.04E-11	1
MSFT	Hybrid	LSTM	-7.75113	9.1E-15	1
PG	Hybrid	ALSTM	-5.03057	4.89E-07	1
PG	Hybrid	CNN	-3.11544	0.001837	1
PG	Hybrid	LSTM	-4.12031	3.78E-05	1
TSLA	Hybrid	ALSTM	0.952265	0.340963	0
TSLA	Hybrid	CNN	-7.63359	2.29E-14	1
TSLA	Hybrid	LSTM	-5.28018	1.29E-07	1
TSM	Hybrid	ALSTM	-9.20645	0	1
TSM	Hybrid	CNN	-4.54568	5.48E-06	1
TSM	Hybrid	LSTM	-4.61378	3.95E-06	1

Table 16. DM Result Comparison of Hybrid, ALSTM, CNN, and LSTM Models.

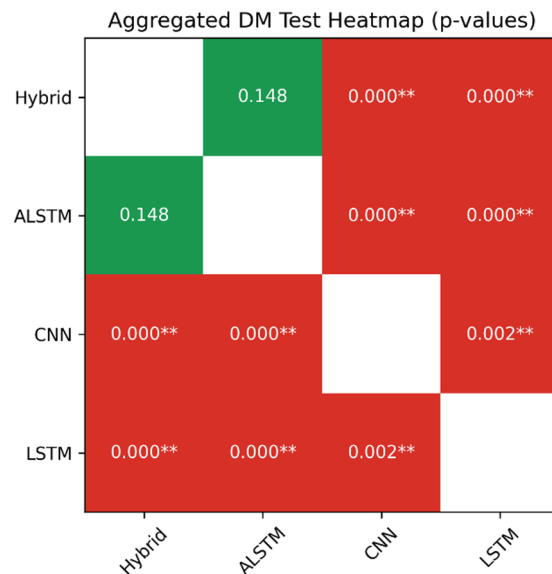


Fig. 19. Aggregated DM test heatmap.

sentiment-aware configuration, the models received ten input features (historical data, technical indicators, and sentiment scores), while in the base configuration without SA, only nine features (historical and technical data) were used. The results showed that the use of SA consistently improved the forecasting performance across most models and stocks. Models with sentiment features calculated lower error measures (MSE, MAE, RMSE, and MAPE) and higher R^2 values, indicating that market sentiment provides valuable behavioral information to the forecasting process. Among all the tested models, the proposed hybrid architecture provided the best overall performance, outperforming the CNN, LSTM, and ALSTM models individually. Its ability to combine spatial feature extraction and time sequence learning enabled it to capture both technical market patterns and psychological dynamics reflected in investor sentiment, thereby providing the most accurate and consistent forecasts.

Theoretical insights

The empirical findings obtained from the eight analyzed stocks provide several important theoretical insights regarding the integration of heterogeneous data sources and hybrid DL architectures in stock price forecasting. The results demonstrate that combining quantitative features (historical and technical indicators) with qualitative sentiment information significantly improves the model's capability to capture the underlying intricacy of financial markets. This improvement can be theoretically explained by the complementarity between rational market indicators and behavioral market signals, where historical and technical data represent observable price dynamics, while sentiment data reflect investors' collective emotions and expectations.

The superior efficiency of the Hybrid model further reinforces the theoretical argument that multi-level feature representation enhances forecasting accuracy. The CNN component effectively extracts spatial and structural dependencies from multidimensional input features, while the ALSTM layer captures long-term temporal dependencies and assigns adaptive attention weights to relevant time steps. This dual mechanism allows the model to balance local trend sensitivity and global sequence understanding, resulting in improved generalization and reduced overfitting compared to single-architecture models such as CNN or LSTM alone. From a theoretical perspective, these findings enhance to the increasing body of literature emphasizing the validity of data fusion and hybrid network design in financial prediction tasks. The evidence from all eight stocks suggests that market behavior is neither purely stochastic nor entirely rational; rather, it emerges from a dynamic interaction between technical signals, historical trends, and investor sentiment. Therefore, the integration of diverse data modalities within hybrid DL architectures offers a more holistic framework for understanding and modeling financial market movements.

Managerial insights

From a managerial and decision-making aspect, the results of this study provide valuable implications for portfolio managers, financial analysts, and corporate decision-makers seeking to enhance the precision and validity of stock price forecasts. The integration of SA into predictive modeling proved to be an effective strategy for improving forecast performance, particularly for stocks with higher volatility and greater sensitivity to investor perception, such as TSLA and AMZN. This finding underscores the managerial importance of monitoring market sentiment and social media signals as part of investment analytics, since emotional and behavioral factors increasingly influence short-term market fluctuations. The consistent superiority of the Hybrid CNN-ALSTM model across multiple stocks also offers key insights for data-driven management and risk assessment. Managers can utilize such hybrid DL frameworks to create dynamic forecasting systems that simultaneously process historical trends, technical indicators, and real-time sentiment data. This capability

enables more adaptive investment strategies, allowing decision-makers to anticipate potential price reversals or volatility surges before they occur.

Moreover, incorporating hybrid models with sentiment-driven features can improve strategic asset allocation and risk diversification, particularly in fast-moving sectors such as technology and consumer goods. These models help reduce uncertainty in prediction-based decisions, supporting more evidence-based portfolio management. In addition, corporate executives may leverage these analytical tools to better understand investor sentiment toward their own firms, which can guide communication strategies, brand positioning, and timing of financial disclosures. In summary, the managerial insight derived from this research is that the integration of behavioral data (sentiment) and hybrid DL models provides a more comprehensive, flexible, and robust decision-support framework. This combination enables managers to move beyond purely historical analyses toward a behavior-aware, data-driven forecasting approach, improving both strategic planning and market responsiveness.

Limitation

In spite of the promising results achieved in this study, several limitations must be acknowledged. First, as with most AI-based forecasting systems, the models used in this research are subject to the phenomenon known as AI hallucination, in which deep learning models may generate seemingly plausible yet potentially misleading predictions based on patterns that do not truly exist in the underlying data. This issue is particularly relevant in financial time-series forecasting, where market noise, abrupt shocks, and non-stationary behavior may cause the model to overfit or overinterpret short-term fluctuations as meaningful signals. Accordingly, the outputs of such models should be interpreted with caution and should ideally be complemented by domain knowledge and expert judgment in real-world investment settings. In addition, the empirical setting of this study is limited in both temporal and geographic scope. The experiments were conducted using a one-year sample period and a small set of eight representative U.S. equities listed on NASDAQ and NYSE. While these markets are highly important, they also possess distinctive microstructure characteristics, including high liquidity, extensive institutional participation, continuous trading, and a relatively mature information environment. Therefore, the findings of this study should be interpreted within the context of large-cap U.S. equities, and they cannot be assumed to automatically generalize to other stock markets with different trading behaviors, investor structures, or information environments.

Second, the study was constrained by data availability and accessibility. Although multiple sources of information historical prices, technical indicators, and textual sentiment were integrated, the accessible dataset was limited to the data available through publicly obtainable sources, including Kaggle-based datasets, which restricted both the sample period and the range of markets examined. In particular, the sentiment component relied on English-language Twitter data, which may not adequately represent investor discourse in non-English-speaking markets or in regions where investors communicate primarily through other social media platforms. As a result, the applicability of the proposed sentiment-analysis framework to markets such as KRX, TSE, Shanghai, or Shenzhen remains unverified in the current study. Nevertheless, this limitation may be addressed in future work by collecting local-language financial news and social media content and, where appropriate, translating such content into English for processing through models such as FinGPT, or alternatively by employing multilingual financial language models directly. Furthermore, the selected sample period, spanning from September 2021 to September 2022, reflects a specific market regime shaped by post-pandemic recovery and the onset of monetary tightening; hence, the robustness of the proposed framework across alternative market conditions, such as prolonged bear markets, financial crises, or periods of extreme volatility, remains to be investigated. Future research with broader international datasets, longer time horizons, and multilingual sentiment sources would therefore be essential for improving the external validity and generalizability of the findings.

Conclusion

In this study, we attempted to present a multimodal model for stock price forecasting. The proposed model includes three input data components, namely textual data, price data, and technical indicators. The FinGPT model is employed for sentiment analysis. The findings of this study provide compelling empirical evidence that integrating sentiment-based features into stock price forecasting frameworks significantly enhances forecasting performance. In all evaluated models, the inclusion of SA resulted in a significant improvement in forecast accuracy, with a reduction in error measures such as MSE, MAE, and RMSE of between 10 and 20%, along with a consistent increase in R^2 . These results confirm that market sentiment acts as a meaningful explanatory variable and complements traditional historical and technical indicators by taking into account behavioral and psychological aspects of investor decision-making that are often neglected in purely quantitative models. Among all configurations, the proposed hybrid model showed the highest level of robustness and accuracy compared to the baseline models. Its architecture effectively captures temporal dependencies and interactions of spatial features, enabling a deeper understanding of nonlinear relationships in financial time series. For example, in the case of TSM, the inclusion of sentiment data reduced the MSE from 11.05 to 8.36 and improved the R^2 from 0.54 to 0.65, while for TSLA, the inclusion of sentiment information reduced the MSE from 445.52 to 339.45, demonstrating the quantitatively tangible benefits of combining multimodal data. Such improvements highlight the superior adaptability of the hybrid structure in learning from diverse and dynamic market signals.

From a theoretical perspective, these findings reinforce the idea that stock market dynamics are not solely governed by quantitative principles, but also by behavioral finance principles, where collective investor sentiment and public perception have a measurable impact on price formation. Thus, the demonstrated success of emotion-enhanced DL architectures bridges the gap between psychological market behavior and data-driven financial modeling. From a managerial and practical perspective, this study emphasizes the strategic importance of incorporating sentiment-based analytics into investment and risk management systems. The ability to gauge

market sentiment through advanced NLP models, such as FinGPT, enables portfolio managers and traders to detect early signals of market optimism or pessimism, optimize asset allocation, and proactively respond to volatility. As a result, the integration of such hybrid intelligence frameworks into financial decision support systems can lead to improved portfolio diversification, reduced exposure to market shocks, and increased profitability.

Several promising directions are identified for future research. The proposed framework can be extended to multi-asset forecasting by incorporating inter-asset dependency structures, enabling the model to capture relational market information alongside temporal and sentiment-based features. In future research, the proposed framework can also be expanded using broader data from international markets, longer time horizons, diverse market regimes, and multilingual news sources and social media, so that it can more accurately assess its generalizability, resilience, and performance in more varied financial environments. Also, future research may improve the proposed framework by using sentiment aggregation based on credibility weighting, adjustments of specific stock technical indicators, and broader evaluations under various market conditions. Furthermore, the inclusion of multi-source sentiment streams such as Reddit discussions, financial news headlines, and corporate announcements may provide a more comprehensive representation of market sentiment. Adopting XAI techniques can also increase interpretability and transparency, addressing a critical limitation of black-box DL models in financial contexts. Finally, extending this framework through crowd learning, transfer learning, or transformer-based architectures may yield more robust, scalable, and adaptive forecasting systems capable of dynamically responding to rapidly evolving market environments.

Data availability

The data acquisition procedures employed in this study fully complied with the terms of service and usage policies established by the respective data providers, namely Yahoo Finance and Kaggle. All datasets were gathered and utilized in accordance with ethical research standards, ensuring that no personally identifiable or sensitive information was involved in the SA process. The collected data was strictly used for academic and non-commercial research purposes. The tweet dataset employed in this study is publicly accessible via Kaggle and can be retrieved from the following link: <https://www.kaggle.com/datasets/equinxx/stock-tweets-for-sentiment-analysis-and-prediction>.

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Author contributions

All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Mobin Golabzaei, Esmaeil Taheripour and Seyed Jafar Sadjadi. The manuscript was written by Mobin Golabzaei, Esmaeil Taheripour and Seyed Jafar Sadjadi. All authors read and approved the final manuscript.

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